mobile-radio: USB-C battery pack with UHF radio

Design document for review: PMR446 + 70 cm FM, LoRa 433/868, APRS and messaging, FreeDV digital voice, USB-C Power Delivery, sound-card profile, ESP32-C5 option

Status	review draft, comments welcome on any numbered chapter / table / requirement ID
Date	2026-11-01
Repository	github.com/sq6emm/mobile-radio, branch ccr-5798edfe-vwv17y, commit 5c36f89
Built with	tools/make_pdf.py (pandoc + Chromium)

Items marked **VERIFY** must be confirmed on hardware. Chapter 15 is a trade study that asks for a decision; chapters 1-14 reflect decisions already taken in the design reviews.

0. Overview

A USB-C battery pack with a built-in UHF radio. It snaps to the back of a phone with a Qi2/MagSafe-style magnet ring, charges the phone over USB-C Power Delivery, and at the same time appears to the phone as a USB device that gives an app push-to-talk voice on the licence-free PMR446 band, and, for licensed amateurs, the 70 cm band with APRS, text messaging, LoRa (433 and 868 MHz) and FreeDV digital voice. A Digirig-compatible sound-card profile lets existing digital-mode programs and iPhones use it without the app. It also works alone, as a power bank and as a basic walkie-talkie.

```
+---------------------------- phone ----------------------------+
|  app: PTT, channel, CTCSS, RSSI, battery, charging control    |
+---------------------------- USB-C ----------------------------+
          | PD: pack = power SOURCE, data = UFP (device)
+---------------------------- pack -----------------------------+
| USB-C PD/charger  <->  2S Li-ion  ->  4.0 V rail -> SA818 UHF |
| STM32F405 (USB device, audio DMA, control)                    |
| SA818 ANT -> 7-pole LPF -> (opt. BPF) -> match -> integral ant |
+---------------------------------------------------------------+
```

What is in this repository

Path	Contents
`docs/01-requirements.md`	Numbered requirements with rationale and acceptance criteria
`docs/02-system-architecture.md`	Block diagram, power budget, interfaces, key design decisions
`docs/03-power-and-usb-pd.md`	Battery, charger, rails, USB-C PD role negotiation
`docs/04-rf-front-end.md`	Radio module, harmonic LPF, optional BPF, coexistence with the phone's radios
`docs/05-antenna.md`	Integral antenna options, matching, ERP budget, range
`docs/06-usb-interface-and-protocol.md`	USB descriptors and the MRP control/audio protocol
`docs/07-audio-path.md`	Analog audio chain and audio filtering
`docs/08-firmware.md`	Firmware architecture and safety interlocks
`docs/09-mechanical-and-thermal.md`	Enclosure, stack-up, heat
`docs/10-regulatory-and-safety.md`	PMR446 rules, RED/EMC/safety standards, battery transport
`docs/11-test-plan.md`	Bring-up, RF conformance, PD interoperability, battery, data/LoRa/DV tests
`docs/12-data-and-aprs.md`	1200 Bd AFSK, AX.25, APRS beacons, text messages, channel arbitration
`docs/13-lora.md`	Two SX1262 modules, antenna sharing, LoRa-APRS, presets, duty limits
`docs/14-digital-voice.md`	FreeDV 2400B / Codec2 through the FM path
`docs/15-esp32-c5-option.md`	ESP32-C5 trade study: wireless phone link, 2.4/5 GHz pack-to-pack
`docs/mobile-radio-design.pdf`	All chapters as one review PDF (`python3 tools/make_pdf.py`)
`hardware/rf/filter_design.py`	LPF/BPF synthesis and lossy simulation (pure Python + numpy)
`hardware/rf/antenna_calc.py`	Helix/IFA sizing, L-match, ERP and link budget
`hardware/rf/results/`	Generated filter tables and antenna report
`hardware/schematics/`	Sheet-by-sheet connection tables for schematic capture
`hardware/bom.csv`	Bill of materials, key parts with alternatives
`firmware/`	Portable C modules (protocol, band plan, radio driver, PD policy, PTT guard, AFSK/AX.25/APRS, messaging, beacons, arbiter, resampler, SX1262 driver, FreeDV glue), application, host link check, unit tests
`firmware/third_party/codec2`	Codec2 / FreeDV library (git submodule, LGPL 2.1)
`tools/mrp.py`, `tools/afsk_wav.py`	Python reference implementation of the protocol and of AX.25/AFSK, bench CLI, WAV generator

Path	Contents
`app/README.md`	Phone-side integration notes (Android USB host, audio, PTT)

Quick start

```
# RF design numbers
python3 hardware/rf/filter_design.py --out hardware/rf/results
python3 hardware/rf/antenna_calc.py report

# firmware unit tests + whole-application link against the host port (gcc only)
git submodule update --init --depth 1   # Codec2 headers
make -C firmware all

# Python implementation cross-checked byte for byte against the C reference vectors
python3 tools/test_mrp.py
python3 tools/afsk_wav.py --src SQ6EMM-7 --info '!5106.50N/01702.25E[test' --out packet.wav
```

Headline design choices

- **Bands and radio**: PMR446 (16 FM channels, 500 mW ERP, integral antenna) for everyone, plus a HAM mode for 430-440 MHz unlocked by a callsign. One SA818U/DRA818U UHF module (1 W / 0.5 W) behind a 7-pole low-pass and a 2-pole 439 MHz band-pass; the helix is tuned to 439 MHz to cover both.
- **LoRa**: two SX1262 modules. 433 MHz shares the helix through an RF switch (LoRa-APRS 433.775, pack-to-pack, ham); 868 MHz has its own printed antenna (pack-to-pack, Meshtastic EU_868 radio settings as a preset). Voice always wins the shared antenna.
- **Data and digital voice**: 1200 Bd AFSK AX.25 with APRS beacons and acknowledged text messages (FM and LoRa), FreeDV 2400B digital voice through the FM path (HAM only). Audio hardware runs at 48 kHz with flat bypass paths so 9600 Bd modes can follow in software.
- **Power**: 2S Li-ion (29.6 Wh), BQ25798 buck-boost charger with OTG output, TPS25751 USB-C PD controller. The pack is the power **source** and the data **device** at the same time, reached through PD role swaps.
- **Phone link**: profile "app" = one vendor USB interface carrying the MRP protocol (control, status, 16 kHz PCM, messages, LoRa, APRS) plus a CDC port; profile "sound-card" = UAC2 48 kHz audio plus CDC with RTS as PTT, Digirig-compatible, for iOS and for Direwolf/FreeDV/SSTV/VARA on any host.
- **Attachment**: Qi2 MPP / MagSafe-compatible magnet ring with orientation magnet, so the pack snaps onto the phone upright (antenna vertical, cable aligned with the port), plus a 100 mm USB-C cable.
- **Safety interlocks in firmware**: host keep-alive, 180 s transmit time-out, thermal and low-battery inhibit, busy-channel lockout, ERP cap from the measured antenna gain.

Status

Design package and verified software building blocks (all unit tests pass, the whole application links against a host stub, Python and C agree byte for byte). No PCB has been fabricated yet; the STM32 port, the KiCad board and the Android app are the next phases. Items that must be confirmed on hardware are marked **VERIFY** throughout the docs. The FreeDV glue is tested against a stand-in; the real Codec2 round trip still has to be built on a developer machine (docs/08, 8.6).

1. Requirements

IDs are stable; later documents refer to them. "Shall" = required for v1, "should" = planned, "may" = optional.

1.1 Use cases

- **UC1 Charged phone with radio**: the pack sits on the back of the phone, tops it up, and the phone app gives push-to-talk voice to other PMR446 radios (any brand) on the chosen channel.
- **UC2 Standalone walkie-talkie**: no phone, or phone asleep. A PTT button, a small speaker and a microphone on the pack work on the last channel.
- **UC3 Plain power bank**: charge the phone, be charged from a USB-C charger.
- **UC4 Group text (phase 2)**: short text messages between packs over the air.

1.2 Radio

ID	Requirement	Rationale / acceptance
R1	Shall operate on the 16 PMR446 analogue channels, 446.00625 + (n-1) x 12.5 kHz MHz, FM, 12.5 kHz channel spacing.	ECC/DEC/(15)05, EN 300 296. Verified with a service monitor.
R2	Shall not exceed 500 mW ERP on any channel, including tolerance and temperature.	Legal limit. Firmware selects the module power level from the measured antenna gain (docs/05).
R3	Shall use an integral (non-detachable, no user-accessible connector) antenna.	Mandatory for PMR446. An internal RF test switch is allowed and provided.
R4	Shall support CTCSS (38 tones) encode/decode and should support DCS.	Interoperability with consumer PMR446 radios.
R5	Shall meet EN 300 296 spurious emission limits: <= -36 dBm below 1 GHz, <= -30 dBm above 1 GHz, conducted at the antenna port.	External 7-pole LPF (docs/04).
R6	Receiver sensitivity should be <= -116 dBm for 12 dB SINAD at the antenna port including filter loss.	Module -124 dBm typical minus ~1-3 dB front-end loss.
R7	Shall keep receiving (desense <= 6 dB) while the attached phone transmits at full power on LTE/5G bands 700 MHz and above, with the pack and phone stacked.	Coexistence is the point of the product. BPF option and shielding (docs/04).
R8	Transmit time-out shall be 180 s with 10 s cool-down; a lost host shall release PTT within 2 s.	Stuck-transmitter protection.
R9	Should provide a busy-channel lockout option.	Courtesy, and required by some national rules for data.

1.2b Amateur (70 cm) mode

ID	Requirement	Rationale / acceptance
R10	Shall offer a HAM mode tuning 430.000-440.000 MHz on a 12.5 kHz grid, FM 12.5/25 kHz, up to 1 W conducted, only after a syntactically valid amateur callsign has been stored.	Licensed users; the licence holder is responsible for legal use (docs/10). `bandplan.c`.
R11	Shall identify with the stored callsign (and SSID) in every AX.25/APRS/LoRa-APRS frame; the app shall remind the operator to identify by voice.	Station identification rules.
R12	Shall provide APRS position beacons (fixed interval or SmartBeaconing) on 432.500 MHz by default (user settable in band) via 1200 Bd AFSK, and on 433.775 MHz via LoRa-APRS.	docs/12, docs/13.
R13	Shall provide text messages between stations using the APRS message format with acknowledgements, over FM AFSK and over LoRa, in both modes (callsign in HAM mode, nickname in PMR mode).	Interoperates with any APRS-capable radio or iGate.
R14	Should provide FreeDV 2400B digital voice (Codec2 1300 bit/s) through the FM path, HAM mode only.	docs/14.

ID	Requirement	Rationale / acceptance
R15	The audio path shall be flat to >= 6 kHz and sampled at 48 kHz so that 9600 Bd and other audio-coupled digital modes can be added in software later.	Keeps the door open (M17, G3RUH, POCSAG RX).

1.2c LoRa

ID	Requirement	Rationale / acceptance
L1	Shall contain a 433 MHz SX1262 LoRa transceiver sharing the helix antenna through an RF switch, and an 868 MHz SX1262 transceiver with its own internal antenna.	User decision; docs/13.
L2	433 MHz LoRa: 433.050-434.790 MHz; <= 10 mW ERP in PMR (ISM) mode, up to +22 dBm conducted in HAM mode.	EN 300 220 vs amateur licence.
L3	868 MHz LoRa: 869.400-869.650 MHz, <= 500 mW ERP, <= 10 % duty cycle (firmware enforced), or 868.0-868.6 MHz <= 25 mW, 1 %.	EN 300 220 Annex B.
L4	Shall transmit and receive LoRa-APRS frames (433.775 MHz, SF12/BW125/CR4-5) and pack-to-pack messages/positions on both bands.	docs/13.
L5	FM voice shall pre-empt any LoRa 433 transmission; LoRa 433 shall be held in standby while the FM transmitter is keyed; the 868 radio is independent.	Shared antenna, receiver protection; `arbiter.c`.
L6	Meshtastic radio settings shall be selectable as a PHY preset; Meshtastic protocol interoperability is a separate later effort.	Honest scope.

1.2d Digital voice

ID	Requirement
D1	FreeDV 2400B TX and RX in firmware (STM32F405 class CPU), speech 8 kHz, modem 48 kHz through the FM audio path.
D2	DV is available only in HAM mode; the app and the pack LED show DV sync state.
D3	Codec2 is LGPL 2.1: the firmware must allow relinking (shipped as a separately updatable object / documented build), see docs/14.

1.3 Power and USB-C

ID	Requirement	Rationale / acceptance
P1	Shall contain a 2S Li-ion battery of about 30 Wh (<= 100 Wh).	Charges a phone ~1.3x; carry-on compliant.
P2	Shall source USB PD 5 V/3 A and 9 V/2 A (18 W) to the phone, and should source 12 V/1.5 A.	Covers phones' normal charge profiles.
P3	Shall sink 5/9/15 V PD at up to 27 W to charge itself, and 5 V/1.5 A from legacy ports.	0.5 C charge of 4 Ah cells.
P4	Shall be the PD power source and USB data device (UFP) simultaneously when attached to a PD phone; shall request the necessary PR_Swap / DR_Swap and fall back gracefully.	USB Type-C/PD roles (docs/03).
P5	Shall stop sourcing power below a configurable reserve (default 20 %) so the radio stays usable; shall resume with hysteresis.	Radio has priority over phone charging.
P6	Shall keep the radio powered from the battery while charging.	Power-path charger.
P7	Radio rail shall be 4.0 V +/- 3 % with < 10 mVpp ripple at the module during RX; TX current step of 0.8 A shall not drop it below 3.6 V.	Module supply spec 3.3-4.5 V; noise into the receiver.
P8	Shall survive USB-C faults: VBUS short, reverse current, 20 V on VBUS while sourcing, CC short to VBUS.	Standard PD source/sink protections.
P9	Battery shall have primary and secondary protection (OV, UV, OC, OT) and a fuel gauge with +/- 5 % SoC.	IEC 62133-2, UN 38.3.

1.4 Phone interface

ID	Requirement	Rationale / acceptance
U1	Shall enumerate as a USB 2.0 full-speed device with a vendor-specific interface carrying the MRP protocol (docs/06).	Works with Android USB Host API, Linux, Windows (WinUSB), macOS.
U2	Shall stream 16 kHz, 16-bit mono PCM in both directions with end-to-end latency < 60 ms (app profile); hardware samples at 48 kHz and firmware resamples.	Natural PTT conversation; R15.
U3	Shall offer a second USB profile ("sound-card" / Digirig-compatible): USB Audio Class 2.0 48 kHz mono in/out plus a CDC-ACM port whose RTS (or DTR) keys PTT. Selected by the app, a long button press or a stored setting; the default profile is "app".	iOS voice, and every PC/phone digital-mode program (Direwolf, FreeDV, SSTV, VARA FM) works unchanged.
U4	Shall report battery state, charging state, PD contract, RSSI, squelch state, PTT state and faults at >= 1 Hz.	App UI.
U5	Shall accept channel, CTCSS/DCS, squelch, power level, volume, filter and charging policy settings.	App UI.
U6	Shall not require a driver on Android or a kernel module on Linux.	Plug and play.

1.5 Mechanical, thermal, safety

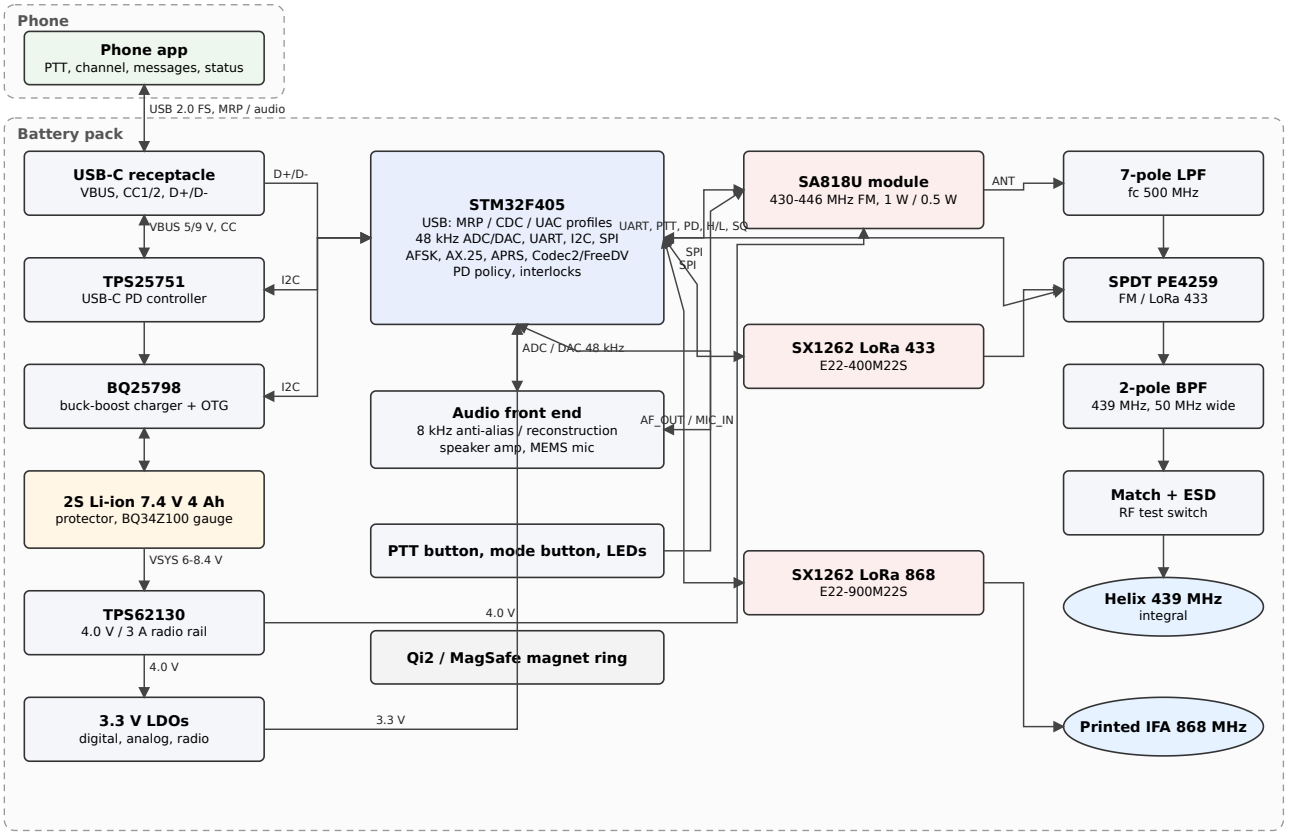
ID	Requirement	Rationale / acceptance
M1	Shall be <= 165 x 78 x 15 mm excluding the antenna stub, with a short USB-C cable to the phone port.	Fits large phones; removable.
M1b	Shall attach to the phone with a magnet ring following the Qi2 Magnetic Power Profile / MagSafe ring geometry, including the orientation magnet, with >= 8 N shear hold.	Snaps onto any ring-equipped phone or case in the correct orientation, keeping the antenna vertical and the cable aligned with the port (docs/09).
M2	Antenna shall be vertically polarised when the phone is held upright.	Other PMR446 radios are vertical; cross-pol costs 10-20 dB.
M3	Surface temperature shall stay < 45 C at 18 W sourcing plus 10 % TX duty at 25 C ambient.	EN 62368-1 touch temperature, user comfort.
M4	Shall meet EN 62368-1 (safety), EN 301 489-1/-5 (EMC), EN 300 296 (radio), EN 50566 / EN 62209 (RF exposure) for CE under RED 2014/53/EU.	Market access (docs/10).
M5	Should be IP52 (dust, dripping) with the USB-C port open.	Outdoor use.

1.6 Out of scope for v1

- Standards-based digital PMR446 (dPMR446 / DMR446: licensed AMBE vocoder), repeaters, higher power, other bands.
- Bluetooth or Wi-Fi links to the phone (USB only).
- Meshtastic protocol stack (routing, encryption, protobufs): only its radio settings are provided as a preset.
- 9600 Bd / M17 / POCSAG: software additions later; hardware keeps the path (R15).
- Qi wireless charging (coil area and EMI conflict with the antenna and receiver).

2. System architecture

2.1 Block diagram



2.2 Design decisions

Decision	Choice	Why	Rejected alternatives
FM radio	SA818U-V4 / DRA818U module	Certified-grade 400-480 MHz FM transceiver with PA, T/R switch, audio processing, CTCSS/DCS, UART control, 1 W, cheap and shielded; covers PMR446 and the 70 cm amateur band.	Discrete RDA1846S + PA (more work, same chip); SA868 2 W (power not usable, hotter).
Modes	PMR446 (anyone) + HAM 430-440 MHz (unlocked by callsign)	Same hardware; firmware band plan and power rules.	Separate products.
LoRa	Two SX1262 modules: 433 MHz on the shared helix via SPDT, 868 MHz on its own printed antenna	No dual-band module exists; two modules are cheaper and safer than a custom dual-band front end; 868 stays concurrent with FM.	Bare SX1262 + custom dual-band matching (phase 2 option); 433 only; 868 only.
Digital voice	FreeDV 2400B / Codec2 1300 in firmware	Open, runs on STM32F405 (SM1000 precedent), passes through the FM audio path. HAM only.	M17 / DMR 4FSK (need a flat 9600 Bd modulator path: verify on the module first), dPMR446 (licensed AMBE).
Sound-card profile	UAC2 48 kHz + CDC RTS-PTT (Digirig-compatible)	Every existing digital-mode program and iOS work without our app.	Vendor-only link.

Decision	Choice	Why	Rejected alternatives
Antenna	Integral stubby helical (v1), PCB IFA option	Vertical, predictable, least affected by the phone body.	Internal-only IFA first (risky when stacked on a metal phone), see docs/05.
Filters	External 7-pole LPF in common path; BPF optional	Module harmonics alone are marginal against -36 dBm; LPF also protects the receiver from the phone's 700-900 MHz uplink.	Elliptic LPF (sharper but less tolerant); SAW in common path (cannot take 1 W).
MCU	STM32F405RGT6 (168 MHz, 192 KB RAM, FPU)	USB FS device, two 12-bit DACs and fast ADCs for 48 kHz audio without a codec, DMA, TinyUSB; Codec2 1300 + FreeDV 2400B are proven on this exact part (SM1000). Risk: the combined load (DV + USB audio + AFSK RX + two LoRa drivers) must be profiled in Phase C; STM32F427/H743 are the drop-in escalation.	RP2040 (no DAC, no FPU), ESP32-S3 (adds a 2.4 GHz radio next to a -120 dBm receiver).
Phone link	Vendor bulk interface + MRP, UAC optional	Avoids Android global audio re-routing, one protocol for everything, works on Linux/Windows for bench.	UAC + HID (Android hijacks audio; iOS has no vendor access anyway); BLE (another radio, latency, pairing).
PD controller	TPS25751D with BQ25798	TI reference pairing for DRP power banks, handles swaps and protections, I2C to MCU.	FUSB302B + software PD stack on the MCU (more flexible, much more firmware; kept as alternative).
Battery	2S1P 4 Ah Li-ion pouch	9/12 V source with buck-boost efficiency, thin, 29.6 Wh.	1S (boost from 3.7 V to 9 V at 2 A is lossy, hot); 2S2P 18650 (too thick for a phone-back pack).
Audio	MCU ADC/DAC at 48 kHz, 8 kHz analog corners, firmware resampling to 16 kHz for voice	Voice needs only 16 kHz, but 48 kHz with a flat 6 kHz path keeps 9600 Bd modes, FreeDV (48 kHz modem) and the UAC sound-card profile possible.	16 kHz only (cheaper filters, closes doors).
Magnetic attachment	Qi2 MPP / MagSafe ring with orientation magnet	Snaps upright on any ring-equipped phone or case.	Clip-on case frame (kept as alternative).
2.4 / 5 GHz (under review, docs/15)	ESP32-C5 as a connectivity co-processor: BLE/Wi-Fi phone link (iOS without MFi, no cable for data), ESP-NOW long-range pack-to-pack	Proposed; footprint on the first board at minimum.	ESP32-C5 as main MCU (no USB OTG: loses the USB link and sound-card profile).

2.3 Power budget

Battery 2S, 7.4 V nominal, 4.0 Ah = 29.6 Wh; usable ~27 Wh.

Mode	Consumers	Power	Endurance (radio only)
Standby, FM RX, LoRa 868 RX, USB idle	module 65 mA @ 4.0 V (0.26 W), LoRa RX 5 mA @ 3.3 V, buck/LDO/MCU ~0.18 W, gauge/PD ~0.02 W	~0.5 W	~54 h
LoRa TX burst (either module)	120 mA @ 3.3 V for 0.3-2.3 s per packet	+0.4 W during bursts	negligible on average
Digital voice TX	FM TX 3.3 W + MCU load ~0.1 W	~3.4 W	as voice
RX with audio to phone	+ ADC/DMA/USB ~0.1 W	~0.55 W	~49 h
TX 1 W RF (module ~750 mA @ 4.0 V)	3.0 W + 0.3 W	~3.3 W	10 % duty average ~0.8 W -> ~34 h

Mode	Consumers	Power	Endurance (radio only)
Sourcing 18 W to phone	18 W out, ~94 % buck-boost	~19.2 W from battery	~1.4 h, delivers ~25 Wh
Charging the pack at 27 W input	2 A into cells	-	0-80 % in ~1.7 h

Radio keeps priority: sourcing stops at 20 % state of charge, leaving about 5.5 Wh, i.e. roughly 7 h of RX/TX use (docs/03).

2.4 Interfaces inside the pack

From	To	Signal
MCU USART1	SA818 RXD/TXD	9600 8N1 AT commands, 3.3 V logic
MCU GPIO	SA818 PTT (active low), PD (power down, low = sleep), H/L (power level)	open-drain or push-pull 3.3 V
SA818 SQ	MCU GPIO (EXTI)	squelch status, low when carrier present (VERIFY polarity on the module revision)
SA818 AF_OUT	buffer + 8 kHz anti-alias filter -> MCU ADC1 at 48 kHz	0-3.3 V audio, DC biased
MCU DAC1 at 48 kHz	8 kHz reconstruction filter -> attenuator -> SA818 MIC_IN	10-100 mVpp into the module mic input (VERIFY level)
MCU I2C1	TPS25751, BQ25798, BQ34Z100-G1	400 kHz, with interrupt lines from PD and charger
MCU SPI2	SX1262 433 and SX1262 868 (separate NSS, BUSY, DIO1, NRST, TXEN/RXEN)	8 MHz
MCU GPIO	PE4259 SPDT control (FM / LoRa433 antenna)	arbiter-driven
MCU USB	profile 0: vendor MRP + CDC; profile 1: UAC2 48 kHz mono in/out + CDC (RTS = PTT)	soft re-enumeration on profile change
MCU USB FS	USB-C D+/D- (both pairs tied)	12 Mbit/s
MCU GPIO	PTT button, 2 LEDs, speaker amp enable, mic enable	-
MCU ADC	PA thermistor, antenna-switch sense (test port inserted)	-

3. Power and USB-C Power Delivery

3.1 Battery

- 2S1P Li-ion pouch, 2 x 4000 mAh, 7.4 V nominal, 8.4 V max, 6.0 V cut-off, 29.6 Wh. Cells with a 0.5 C (2 A) charge rating and 2 C discharge (8 A), e.g. 606090-class pouch cells 70 x 60 x 6 mm.
- **Primary protection**: 2S protector IC (Seiko S-8252 class or equivalent) driving dual N-channel FETs in the low side: OV 4.25 V/cell, UV 2.8 V/cell, OC discharge 8 A, short-circuit, with cell balancing provided by the charger (BQ25798 has none; use a 2S balancer front end such as BQ29209 or the gauge's balancing if a BQ40Z50 class gauge is chosen).
- **Secondary protection**: BQ7718 over-voltage protector blowing a 2S chemical fuse (Dexerials SFJ class) at 4.35 V/cell.
- **Fuel gauge**: BQ34Z100-G1 (Impedance Track, multi-cell, I2C), 5 mOhm sense resistor, NTC on the cells. Alternative: estimate from BQ25798 ADC + coulomb counting in firmware (cheaper, +/-10 %).
- NTC thresholds: charge only between 0 and 45 C, discharge -20 to 60 C, TX inhibit above 60 C cell temperature.

3.2 Charger and power path: BQ25798

Buck-boost charger, 1-4S, input 3.6-24 V, 5 A charge, integrated OTG output 2.8-22 V up to 3.3 A (datasheet values; **VERIFY** OTG current at 9 V on the chosen revision, the 18 W source PDO assumes 2 A at 9 V).

- Battery side: 2S, charge voltage 8.4 V, charge current 2.0 A, termination 200 mA, pre-charge 300 mA, safety timer 8 h.
- VSYS: tracks the battery (6.0-8.4 V) with 6.2 V minimum system voltage, so the radio keeps running from VBUS even with an empty battery (P6).
- Input current limit set from the PD contract by the MCU (ILIM_HIZ pin tied to TPS25751 is also possible, but register control is simpler to debug).
- OTG mode: the MCU writes the OTG voltage (5.0/9.0/12.0 V) and current limit when the PD controller reports a source contract.
- Inductor 2.2 uH / 8 A saturation, shielded; input and output 4 x 22 uF 25 V X7R. Switching frequency 750 kHz with dithering **off** during reception if spur tests show problems (1.5 MHz option moves harmonics; the 446 MHz receiver is 595th harmonic away, well below any realistic harmonic amplitude, but the layout must keep the switch node tiny).

3.3 Rails

Rail	Source	Load	Notes
VSYS 6.0-8.4 V	BQ25798	buck, LDOs, speaker amp	main bus
+4V0 / 3 A	TPS62130 (adjustable sync buck, 2.5 MHz, forced PWM)	SA818 VCC	LC post filter 1 uH + 22 uF, then 100 uF low-ESR at the module; ripple < 10 mVpp. Forced PWM keeps the frequency fixed and out of the audio band in RX.
+3V3D / 300 mA	TPS7A20 from +4V0	MCU, PD, gauge logic, LEDs	low-noise LDO from the 4 V rail (0.7 V drop, 0.2 W worst case)
+3V3A / 150 mA	LP5907 from +4V0	op-amps, mic bias, ADC/DAC reference	separate star ground return to the ADC ground
VBUS	BQ25798 OTG or external	USB-C	5-20 V

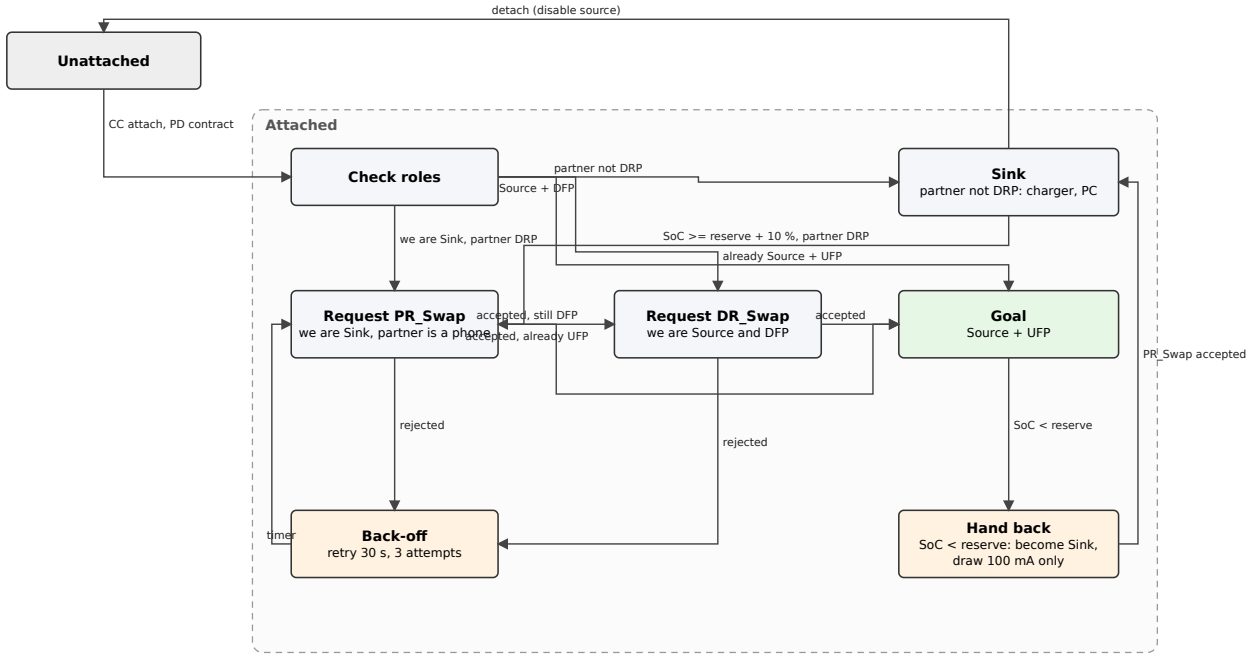
TX current step: the module pulls ~0.8 A within ~1 ms when keyed. With the TPS62130 (3 A) the +4V0 rail dips < 100 mV with 100 uF local bulk; measured in test T-P4 (docs/11).

3.4 USB-C and PD roles: why swaps are needed

Type-C by itself ties the roles together: whoever supplies VBUS is also the USB host (DFP). The pack wants the opposite pairing:

	Power	**Data**
Phone	Sink (gets charged)	Host / DFP (runs the app)
Pack	**Source**	**Device / UFP**

USB PD allows it: after the initial Type-C attach, the roles can be changed independently with `PR_Swap` (power) and `DR_Swap` (data). The pack's policy, implemented in `firmware/src/pd_policy.c` and tested in `firmware/tests/test_pd_policy.c`:



Practical notes:

- Configure the TPS25751 as DRP with **Try.SRC**. Phones implement Try.SNK, so the pack usually wins as Source/DFP, then only a `DR_Swap` is needed; that is the same sequence a USB-C monitor with power delivery uses with a laptop, and phones accept it.
- If the phone ends up Source/DFP (some phones with a full battery refuse Try.SRC), the pack asks for `PR_Swap`. Android accepts it when the pack advertises a usable Source PDO (5 V/3 A, 9 V/2 A).
- While we are the sink of a DRP partner we draw only 100 mA (trickle for the logic), so a phone is never drained by the pack unless the user allows it (`SET_POWER` flag `charge`).
- No PD (5 V legacy host, A-to-C cable): the pack is a plain USB device powered by the phone's OTG 5 V; the radio still works (phone drains ~0.5 W). Power sharing is not possible without PD, the app shows this.
- Wall charger (not DRP): sink at the full negotiated power, no swaps.
- Source PDOs advertised: 5 V/3 A, 9 V/2 A (and 12 V/1.5 A when the measured OTG current allows). Sink PDOs: 5 V/3 A, 9 V/3 A, 15 V/2 A.
- The data lines are 12 Mbit/s full speed only: no SuperSpeed mux, D+/D- of both connector orientations are simply tied together.
- VCONN is not needed (no e-marker required under 3 A / 60 W). The TPS25751 still supplies VCONN if a cable asks.

3.5 Protections

- VBUS: TPS25751 integrated sink path and external source path FETs with OVP (23 V), reverse-current blocking, 5 A short-circuit, discharge resistor.
- CC pins: 24 V tolerant (CC short to VBUS test), ESD 8 kV.

- D+/D-: TVS array (0.5 pF) at the connector, common-mode choke 90 ohm @ 100 MHz, then 22 ohm series resistors at the MCU.
- Thermal: NTC on PA area and charger; firmware throttles source current at 60 C board temperature, inhibits TX at 70 C PA temperature.
- Radio rail enable is sequenced after +3V3D so that the module never sees floating UART/PTT lines (PTT floating can key the transmitter).

4. RF front end and filters

4.1 Radio module

SA818U-V4 (NiceRF) or DRA818U (Dorji), 400-480 MHz FM transceiver module:

Parameter	Value (datasheet, VERIFY per revision)
Supply	3.3-4.5 V, design uses 4.0 V
RF output	1 W (H) / 0.5 W (L), selected by the H/L pin
TX current	<= 750 mA at 1 W
RX current	~60-70 mA; power-down < 1 uA via PD pin
Sensitivity	-124 dBm (12 dB SINAD)
Channel spacing	12.5 / 25 kHz, 12.5 kHz used
Control	UART 9600 8N1, `AT+DMOSETGROUP`, `AT+DMOSETVOLUME`, `AT+SETFILTER`, `AT+SETTAIL`, `RSSI?`
Audio	MIC_IN (electret level), AF_OUT (line level, needs amplifier), built-in pre/de-emphasis, 300 Hz HPF, 3 kHz LPF, CTCSS/DCS
Antenna port	50 ohm pin, internal T/R switch and basic harmonic filter

The module is a complete radio and covers both operating modes (PMR446 at 446 MHz and the 70 cm amateur band 430-440 MHz); everything in this chapter is about making its single antenna pin legal, clean and robust next to a phone, and about sharing the antenna with the 433 MHz LoRa module.

4.2 Signal chain

```
SA818 ANT -> C_dc -> [LPF: C1 L2 C3 L4 C5 L6 C7] --\
                                                    [SPDT PE4259] -> [BPF 2-pole 439 MHz, 50 MHz wide]
->
LoRa 433 module ANT -> [shunt PIN limiter pad] ----/      [RF test switch MM8130] -> [match] -> [ESD] -
> helix

LoRa 868 module ANT -> [10 pF series, shunt L pad] -> printed IFA (own antenna)
```

The FM path and the 433 MHz LoRa module share the helix through the switch; the arbiter (`firmware/src/arbiter.c`) points it at whoever transmits, and at the preferred receiver otherwise (FM by default). The 868 MHz LoRa module is fully independent (docs/13).

- 50 ohm coplanar waveguide with ground, 0.3 mm wide / 0.2 mm gap on 0.2 mm prepreg to L2 (adjust to stack-up), ground vias every 2 mm along the line.
- All filter parts 0603 (inductors) / 0402 C0G (capacitors), inductor Q >= 60 at 450 MHz (wirewound ceramic, e.g. Coilcraft 0603HP / Murata LQW18 class).
- RF test switch (Murata MM8130-2600): inserting a probe disconnects the antenna and gives a conducted port for all compliance measurements. It is under the back cover, not user accessible, so the antenna remains "integral" in the regulatory sense.

4.3 TX harmonic low-pass filter

Synthesised and simulated by `hardware/rf/filter_design.py` (results in `hardware/rf/results/filter_summary.md`).

Design: 7th-order Chebyshev, 0.1 dB ripple, shunt-C first, f_c = 500 MHz, Z0 = 50 ohm.

Element	Ideal	E24 part	Note
C1, C7	7.52 pF	7.5 pF C0G 0402	

Element	Ideal	E24 part	Note
L2, L6	22.6 nH	22 nH 0603 wirewound	
C3, C5	13.35 pF	13 pF C0G 0402	
L4	25.0 nH	24 nH 0603 wirewound	

Simulated with Q_L = 60, Q_C = 300, 0.1 pF inductor parasitic capacitance and 0.4 nH capacitor ESL:

Frequency	S21	Comment
446.0-446.2 MHz	-0.8 dB	passband, S11 -16 dB
703 MHz (LTE B28 UL)	-35 dB	receiver protection
832 MHz (LTE B20 UL)	-50 dB	
892 MHz (2nd harmonic)	-56 dB	with module's own ~ -40 dBc: ~ -96 dBc, limit -66 dBc
1338 MHz (3rd harmonic)	< -70 dB (layout floor)	
>= 1.7 GHz	< -70 dB (layout floor)	

Why 500 MHz and not closer to the band: component tolerance (5 % inductors move the ripple edge by ~2.5 %) and the module's output impedance is not an ideal 50 ohm; a 12 % guard band keeps the passband loss and match stable. A cutoff of 540 MHz would save 0.2 dB and lose 7 dB at 700 MHz; the receiver protection is worth more here.

Spurious budget at the antenna port (EN 300 296, TX): carrier +30 dBm, limit -36 dBm below 1 GHz (-66 dBc) and -30 dBm above 1 GHz (-60 dBc). Module 2nd harmonic typically -40 to -50 dBc, 3rd -50 to -60 dBc. After the LPF both are below -90 dBc, limited in practice by shielding and layout (~ -70 to -80 dBc). Margin > 10 dB.

4.4 Band-pass filter (receiver selectivity), 439 MHz

Purpose: the attached phone transmits up to +23 dBm at 700-960 MHz and 1.7-2.7 GHz, 1-3 cm from the pack antenna. Coupling of -20 to -30 dB puts -7 to +3 dBm into the pack's antenna; the LPF leaves -42 to -32 dBm at 703 MHz, which is near the module's blocking limit (R7). A 446 MHz band-pass adds 30 dB where it matters, and also attenuates TETRA (380 MHz) and LPD/ISM 433 MHz traffic that the LPF passes unfiltered.

With the 70 cm amateur mode and LoRa-APRS at 433.775 MHz the filter must pass 430-446.2 MHz, so it is centred at 439 MHz and widened to 50 MHz (`filter_design.py --bpf-f0 439e6 --bpf-bw 50e6`): 2-pole Butterworth, capacitively coupled shunt resonators, L = 12 nH:

Element	Value
C_end (in/out, series)	2.7 pF
C_couple	0.95 pF
C_res (each resonator, shunt)	7.3 pF (+ 2.2-3.3 pF trimmer range on the pads)
L_res	12 nH, Q >= 120 (air-core 4 turns 1 mm wire on 3 mm, or ceramic coax resonator)

Simulated (Q_u = 120): 1.4 dB at 446.1 MHz, 1.2 dB at 433.9 MHz, -18 dB at 380 MHz (TETRA), -26 dB at 703 MHz, -29 dB at 900 MHz; with the LPF in cascade the cellular bands are below -60 dB. A dielectric 2-pole filter brings the loss to ~0.6 dB.

Caveats that decide the stuffing option:

1. In the common TX/RX path the resonators carry ~1.6 A RMS circulating current at 1 W (Q_L = 11). Chip inductors cannot do that; air-core coils or a commercial 2-pole dielectric filter for 446 MHz PMR (standard part in PMR446 handsets) are required.
2. Alternatively the BPF (or a 446 MHz SAW filter, which can not take TX power) goes in a **receive-only branch** behind an SPDT pair (PE4259, +34 dBm), switched by the PTT line. That costs 0.7 dB in both directions and two more parts but gives the best receiver. The PCB has footprints for both;

option A (common-path dielectric filter) is the default, option B is populated if the coexistence test T-R7 fails.

4.5 Coexistence with the phone and the pack's own electronics

Threats to the -120 dBm receiver:

Source	Mechanism	Mitigation
Phone cellular TX 700-960 MHz, +23 dBm	front-end blocking, reciprocal mixing	LPF (35-50 dB), BPF (26-29 dB), antenna at the pack top edge away from the phone's antennas (usually bottom and top edges: keep the pack antenna off-centre, top-left)
LoRa 868 TX (+22 dBm) 30 mm from the helix	869 MHz is 1.98 x 439 MHz: just below the FM receiver's 2nd harmonic image; direct blocking	LPF + BPF give > 60 dB at 869 MHz; the 868 module's own harmonic filter; firmware may blank FM audio for the 0.3-1 s burst if T-L3 shows a click
FM TX 1 W into the LoRa 433 input through the SPDT	25 dB switch isolation leaves ~+5 dBm at the SX1262 (abs. max +10 dBm)	LoRa 433 in standby during FM TX; shunt PIN limiter footprint; T-L2
LoRa 433 TX (+22 dBm)	into the SA818 through the switch's off port: -3 dBm	SA818 input tolerates it (handhelds see this from nearby radios); FM audio muted while LoRa 433 owns the antenna anyway
Phone 1.7-2.7 GHz, 3.5 GHz, Wi-Fi, BT	out-of-band blocking, PA harmonics landing in band? (none of 446 MHz x n falls in these; the reverse, phone TX / n, does not land on 446 either)	LPF > 70 dB; metal can over the module
Phone display/CPU broadband noise, Qi coil	near-field pickup by the pack antenna and PCB traces	full ground plane facing the phone, shield cans, antenna on the far edge
Pack buck converters (750 kHz, 2.5 MHz)	harmonics at 446 MHz (harmonic #595 / #178) and conducted ripple into the module supply	forced PWM, tiny switch loops, LC post filter, ferrite bead + 100 uF at the module, spread spectrum off during RX
USB 12 MHz data, 48 MHz MCU, 168 MHz core	harmonics; 446 MHz = 37th of 12 MHz, close to 9th of 48 MHz (432) and 3rd of 168 MHz (504)...	pick MCU clock so no harmonic lands in 446.0-446.2 MHz (168 MHz: 2.65th, fine; USB 12 MHz 37th = 444 MHz, within 2 MHz: keep D+/D- short, CM choke, ground stitching); listen for birdies in test T-R6 and move the PLL if needed
Pack TX 1 W into the phone	phone's cellular/GNSS receivers blocked	2nd harmonic at 892 MHz sits in the GSM-900 uplink (not the phone's RX); 4th at 1784 MHz is inside LTE band 3 uplink (also not the phone's RX). The LPF keeps them < -60 dBm. GNSS 1575 MHz is 3.53x, no harmonic. PMR446 TX at 1 W is comparable to the phone's own TX; expect no issue but test T-R8

Shielding: the SA818 comes in a tin can. Add a second can over the LPF/BPF/match area and a ground-stitched fence between the power section and the RF area. The back cover that faces the phone carries a ground-connected copper layer (inner PCB layer) to form a near-complete Faraday box, with only the antenna feed leaving it.

4.6 Power and digital filtering on the RF side

- +4V0 to module: 1 uH + 22 uF LC, then 600 ohm @ 100 MHz ferrite bead + 100 uF + 100 nF + 1 nF at the VCC pin.
- PTT, PD, H/L, SQ, UART: 1 kOhm series + 100 pF to ground at the module (low-pass, keeps the 446 MHz field off the MCU and MCU noise off the module).
- MIC_IN: 100 ohm + 1 nF RF bypass right at the pin; AF_OUT: 1 nF bypass.
- USB D+/D-: common-mode choke, 0.5 pF TVS, 22 ohm series.
- VBUS: pi filter 10 uF / 1 uH / 10 uF, 2 kV-rated ferrite bead on the shell.

5. Antenna

Constraints: PMR446 requires an integral antenna (no user connector), ERP <= 500 mW, vertical polarisation in use (M2), and the antenna sits on a pack that is stuck to a phone with a magnet ring. The same helix now also serves the 70 cm amateur mode (430-440 MHz) and the 433 MHz LoRa module, so it is tuned to **439 MHz**, the geometric centre of 432.5 and 446.1 MHz. A second, printed antenna serves the 868 MHz LoRa module. Numbers below come from `hardware/rf/antenna_calc.py report` (output saved in `hardware/rf/results/antenna_report.txt`).

Wavelength at 439 MHz: 683 mm; lambda/4 = 171 mm. Nothing phone-sized is resonant without loading, so every option is electrically short.

5.1 Options

Option	Description	Expected gain	Pros	Cons
A (v1 baseline)	Fixed stubby normal-mode helix, 60 mm long, 6 mm diameter, ~16 turns, in a TPU boot at the pack's top-left corner, feed through the PCB top edge	-3 to -6 dBd	Vertical, well understood, least sensitive to the phone body, easy to retune by trimming	Sticks out 60 mm; mechanical stress at the base
B (phase 2)	Meandered inverted-F printed along the top 12 mm of the PCB, 60 x 10 mm keep-out, ~151 mm conductor, 10 meanders at 6 mm pitch	-4 to -8 dBd on the phone	Nothing protrudes	Detuned by the phone body and hand (10-20 MHz shifts), needs a tuner network per phone size; efficiency maybe 20-30 %
C	90 mm fold-out whip (PMR446 handheld style)	-1 to -3 dBd	Best range	Fragile on a phone-back product, still must be non-removable

Decision: A for v1 with footprints for B's matching network so that the same PCB can be evaluated with an internal antenna.

Mechanical placement (see docs/09): top-left corner, 25 mm from the magnet ring edge, counterpoise = the pack's PCB ground plane (155 x 70 mm, i.e. about lambda/4 long, which helps: the chassis is half of the radiator). The helix axis stays vertical when the phone is portrait; in landscape the polarisation loss to other handhelds is 10-20 dB, same as any handheld laid down.

5.2 Helix starting dimensions (439 MHz)

Diameter	Axial length	Wire length	Pitch	Turns
6 mm	60 mm	~307 mm (0.45 lambda)	3.75 mm	16.0
8 mm	50 mm	~307 mm	4.1 mm	12.1
5 mm	80 mm	~307 mm	4.2 mm	18.9

Bandwidth: a 60 mm helix on this ground plane has a VSWR < 2 bandwidth of roughly 15-20 MHz. Centred at 439 MHz that covers 432-446 MHz at VSWR <= 2 (about 0.5 dB mismatch loss at either edge). If the measured bandwidth is narrower, the compromise is to favour 446 (the certified mode) and accept ~1 dB at 432.5 MHz; a longer (80 mm) or thicker helix widens the band.

Wire: 1.0 mm phosphor bronze or stainless spring wire on a nylon former, soldered to a brass base insert that is screwed and soldered to the PCB. The k = 0.45 wire-length factor is a starting point; the sleeve dielectric and the pack shift resonance by typically -5 to -10 %, so wind 10 % longer and trim on a VNA with the pack assembled, cover on, on a phone.

Feed impedance of such a stub on a small ground is 15-30 ohm with some reactance. The matching footprint is a pi: shunt L (also the DC/ESD path), series C, shunt C. Example from the calculator for a measured 22 - j15 ohm: series 14.2 nH + shunt 8.1 pF, or series 36 pF + shunt 15.8 nH. Use the low-pass form (series L, shunt C) because it adds harmonic rejection.

5.3 Internal IFA (option B) design notes

- Keep-out: no copper on any layer under the element; ground plane edge is the element's ground reference; feed via 50 ohm CPW from the match network.
- Shorting pin 4-6 mm from the feed; tune resonance with the meander length (longer = lower) and the impedance with the pin distance.
- Simulate in openEMS or NEC with the phone modelled as a 150 x 72 x 8 mm metal box 6 mm behind the pack, hand as a lossy dielectric block; expect -4 to -8 dBd realised gain.
- The magnet ring (steel/NdFeB, ~56 mm diameter) is a large conductor in the near field: keep the element >= 20 mm away from it.

5.3b 868 MHz printed inverted-F (LoRa 868)

lambda/4 at 869.5 MHz is 86 mm, so a printed IFA fits comfortably in a 40 x 8 mm keep-out at the top-right corner (`antenna_calc.py ifa --f 869e6 --width 40 --height 8`: ~78 mm conductor, 5 meanders at 8 mm pitch). Feed via a series 10 pF pad and a shunt inductor pad for tuning; shorting pin 4 mm from the feed. Keep it >= 30 mm from the helix base and >= 20 mm from the magnet ring; expected gain -3 to -5 dBi on the phone. ERP at +22 dBm conducted is then ~60-80 mW, far below the 500 mW allowed in 869.4-869.65 MHz; the firmware's power cap for the 868 band is therefore not antenna critical.

5.4 ERP budget and power level selection

ERP = P_conducted - front-end loss + antenna gain (dBd). Front-end loss is 0.8 dB (LPF) + 0.3 dB (traces, ESD, test switch) = 1.1 dB.

Module power	Antenna gain -2 dBd	-4 dBd	-6 dBd
1.0 W (H)	0.49 W ERP, at the limit	0.31 W	0.19 W
0.5 W (L)	0.24 W	0.15 W	0.10 W

The production test measures ERP (or antenna gain by substitution) and stores the allowed power level per unit (`ANTENNA_GAIN_DBD_X10` in `app_main.c`, later a calibration record). In PMR446 mode the firmware refuses the H setting when the stored gain would put ERP above 26.5 dBm (0.45 W, 0.5 dB margin for temperature and supply); in HAM mode 1 W is always allowed (`bandplan_high_power_allowed`). R2 is thereby met by construction. The same stored gain caps the 433 MHz LoRa power at 10 mW ERP in PMR/ISM mode.

5.5 Range expectation

ERP 0.3 W, receive gain -4 dBi, sensitivity -118 dBm at the port, 10 dB fade margin, 3 dB body loss each end: allowed path loss 125 dB.

Environment	Path-loss exponent	Range
Line of sight (hill to hill, water)	2.0	tens of km (limited by horizon)
Open rural, ground reflections	2.7	~5 km
Suburban	3.2	~1.3 km
Dense urban	3.8	~400 m
Indoors through floors	4.5	~160 m
Plane-earth two-ray, 1.5 m antennas	-	~2 km

These match what consumer PMR446 handhelds achieve, which is the goal: the pack talks to any PMR446 radio.

5.6 RF exposure

A 0.5 W ERP transmitter on the back of a phone is used like a walkie-talkie (held 5-10 cm from the mouth, or on a table with the app), not at the ear. EN 50566 / EN 62209-2 SAR assessment is needed because the device can be within 20 cm of the body; PMR446 handhelds at this power routinely pass. The app instructs speaker-mode use and the firmware's 180 s time-out limits duty cycle. Keep the antenna at the top edge so it is as far as possible from the hand holding the phone.

6. USB interface and the MRP protocol

6.1 Enumeration: two profiles

USB 2.0 full speed (12 Mbit/s), bus-powered descriptor (`bMaxPower` 100 mA, the pack really runs from its own battery) so that no host ever has to supply more than it would to a mouse. The pack has two USB personalities and re-enumerates (soft disconnect) when switching; the choice is persisted.

Profile 0 "app" (default): for the phone app and bench tools.

#	Interface	Class	Endpoints	Purpose
0	MRP	0xFF vendor, subclass 0x01, protocol 0x01	EP1 OUT bulk 64 B, EP1 IN bulk 64 B	control, status, PCM audio, messages, LoRa, APRS (this document)
1-2	Serial	CDC-ACM	EP2 INT, EP3 bulk in/out	debug console / AT pass-through; RTS also keys PTT so Direwolf-style programs work even in this profile

Profile 1 "sound-card" (Digirig-compatible): for iOS and for existing digital-mode programs (Direwolf, FreeDV GUI, SSTV, VARA FM, JS8Call ...).

#	Interface	Class	Endpoints	Purpose
0-2	Audio	UAC 2.0, 48 kHz mono s16 in (receiver) and out (transmitter), asynchronous with feedback	isochronous	the radio's audio, flat path, no processing
3-4	Serial	CDC-ACM	EP INT, bulk in/out	**RTS = PTT** (DTR optional), same as a Digirig; the console is still available

Switching: `SET_PROFILE` over MRP, a long press of the mode button, or the stored setting at power-up. The pack's own PTT button and channel memory keep working in either profile, so an iPhone user talks with the button and listens through the "headset".

- VID/PID: development uses pid.codes 0x1209 / 0x0001 (test allocation); a real PID must be obtained before any unit leaves the bench. The two profiles use different PIDs so operating systems cache the right driver.
- Microsoft OS 2.0 descriptors bind WinUSB to interface 0 automatically.
- BOS also carries a WebUSB platform descriptor so a browser-based bench tool can talk to it; not needed by the phone app.
- Strings: manufacturer, product "mobile-radio PMR446 pack", serial = unique from the MCU UID (also reported in `INFO`).

Why no USB Audio Class by default: Android treats any UAC device as the phone's audio output and reroutes ringtones, media and calls to it the moment it is plugged in. For a PTT accessory that is wrong. Streaming PCM over the vendor interface keeps the phone's audio untouched; the app plays and records with `AudioTrack` / `AudioRecord` as it chooses. UAC remains as a profile for hosts without an app (iOS), where the pack's own PTT button and channel memory are used.

6.2 Frame format

```
offset  size  field
0       1     SOF = 0xA5
1       1     type
2       1     seq        (per-direction counter, wraps; audio uses it for loss detection)
3       2     len        little-endian payload length, 0..1024
5       1     HCS        type ^ seq ^ len_lo ^ len_hi
6       len   payload
6+len   2     CRC-16/CCITT-FALSE (poly 0x1021, init 0xFFFF) over bytes 1..5+len, little-endian
```

Frames are packed back to back in bulk transfers; a transfer may contain several frames and a frame may span transfers. Receivers resynchronise on SOF + HCS and verify the CRC. Reference implementations:

`firmware/src/mrp.c` and `tools/mrp.py`. `firmware/build/vectors` (built by `make -C firmware test`) prints reference frames, an AX.25 frame, its HDLC bit stream and the first AFSK samples; `tools/test_mrp.py` regenerates all of them in Python and asserts byte equality.

6.3 Messages

Host to device:

Type	Name	Payload	Reply
0x01	GET_INFO	-	INFO
0x02	SET_CHANNEL	channel u8 (1-16), bandwidth u8 (0), ctcss_tx u8 (0-38), ctcss_rx u8, squelch u8 (0-8), power u8 (0 low / 1 high), flags u8 (bit0 busy-channel lockout)	ACK
0x03	PTT	state u8, ttl_ms u16 (keep-alive window, 0 = 2000)	ACK (result BUSY / INHIBITED when refused)
0x04	SET_AUDIO	volume u8 (1-8), mic_gain u8 (0-15), filter_flags u8 (bit0 bypass emphasis, bit1 bypass 300 Hz HPF, bit2 bypass 3 kHz LPF)	ACK
0x05	SET_POWER	flags u8 (bit0 allow sourcing to phone, bit1 allow charging from port), max_source_w u8, reserve_soc_pct u8	ACK
0x06	DATA_TX	up to 200 bytes of application data (phase 2, AFSK)	ACK
0x07	GET_STATUS	-	STATUS
0x08	KEEPALIVE	-	-
0x09	SET_MODE	mode u8 (0 PMR446, 1 HAM 70 cm)	ACK (INHIBITED when no callsign stored)
0x0A	SET_FREQ	hz u32, squelch u8, power u8, ctcss_tx u8, ctcss_rx u8, bandwidth u8 (HAM mode only, 430-440 MHz on 12.5 kHz)	ACK
0x0B	SET_IDENTITY	callsign char[8], ssid u8, nickname char[8] (empty = keep); a valid callsign unlocks HAM mode	ACK (detail = unlocked)
0x0C	SET_BEACON	enabled u8, smart u8, interval_s u16, sym_table u8, sym_code u8, transport u8 (0 FM, 1 LoRa433, 2 LoRa868, 3 FM+LoRa433), path char[16], comment char[32]	ACK
0x0D	POSITION	lat_e7 i32, lon_e7 i32, alt_m i16, speed_kmh u16, course_deg i16 (-1 none), valid u8	-
0x0E	SEND_MSG	transport u8, to char[10], text 1-67 bytes	ACK (detail = message id)
0x0F	SET_PROFILE	profile u8 (0 app, 1 sound-card)	ACK, then re-enumeration
0x10	AUDIO_TX	640 bytes s16le PCM, 16 kHz, 20 ms	-
0x11	LORA_CONFIG	radio u8 (0 = 433, 1 = 868), preset u8 (0 custom, 1 LoRa-APRS 433, 2 Meshtastic EU_868 PHY, 3 pack 433, 4 pack 868), freq_hz u32, sf u8, bw u8 (4 = 125 k, 5 = 250 k, 6 = 500 k), cr u8 (1-4), power_dbm i8, sync u16, rx_pref u8 (433: prefer LoRa receiver over FM when idle)	ACK
0x12	LORA_TX	radio u8, raw payload 1-255 bytes	ACK
0x13	SET_DV	enabled u8 (FreeDV 2400B; HAM mode only)	ACK

Device to host:

Type	Name	Payload
0x81	INFO	proto u8, fw_major u8, fw_minor u8, hw_rev u8, caps u32 (bit0 voice, bit1 data, bit2 PD source, bit3 PD sink, bit4 UAC profile), serial char[16], region u8 (0 = EU PMR446), n_channels u8
0x82	ACK	req_type u8, result u8 (0 OK, 1 BAD_LEN, 2 BAD_VALUE, 3 RADIO, 4 BUSY, 5 INHIBITED, 6 UNSUPPORTED), detail u16
0x90	AUDIO_RX	640 bytes s16le PCM

Type	Name	Payload
0x91	STATUS	flags u16 (bit0 PTT, 1 squelch open, 2 TX inhibited, 3 charging, 4 sourcing, 5 fault, 6 radio ok, 7 host alive), rssi_dbm i8, channel u8, batt_mv u16, batt_ma i16, soc_pct u8, temp_c i8, vbus_mv u16, vbus_ma i16 (+ = sourcing), pd_role u8 (bit0 source, 1 DFP, 2 contract, 3 partner DRP), pd_mv u16, pd_ma u16, tx_remaining_s u16
0x92	DATA_RX	raw AX.25 frame as received on FM (for logging / KISS-style apps)
0x93	MSG_RX	transport u8, from char[10], id char[6], text
0x94	MSG_STATUS	id u16, state u8 (1 pending, 2 sent waiting ack, 3 acked, 4 failed)
0x95	LORA_RX	radio u8, rssi_dbm i16, snr_db i8, raw payload
0x96	APRS_RX	transport u8, src char[10], kind u8 (1 position, 5 status), lat_e7 i32, lon_e7 i32, course i16, speed_kmh u16, comment/text
0x7F	NAK	same layout as ACK, for malformed frames

STATUS is 29 bytes: the 21 bytes above followed by mode u8, profile u8, dv_flags u8 (bit0 enabled, bit1 in sync), lora_flags u8 (bit0 433 listening, bit1 433 TX, bit2 868 listening, bit3 868 TX), freq_hz u32.

Rules:

- Every frame from the host refreshes the keep-alive. While PTT is keyed the host must send something (AUDIO_TX frames do) at least every `ttl_ms`, default 2 s, or the transmitter is released (R8).
- STATUS is pushed at 1 Hz and immediately on PTT/squelch/fault changes.
- AUDIO_RX is sent only while the squelch is open (or always, if the app sets squelch 0), so an idle link carries 21 bytes per second.
- Audio frames carry no timestamps; the 20 ms cadence and `seq` are enough for a jitter buffer of 2-3 frames on the phone (40-60 ms, U2).
- SET_CHANNEL is refused with BUSY while transmitting.

6.4 Bandwidth and latency

PCM 16 kHz x 16 bit = 256 kbit/s per direction plus 1.3 % framing; full-speed bulk delivers > 8 Mbit/s, so the bus is never the bottleneck. Latency budget phone-mic to RF: 20 ms frame + USB < 2 ms + DAC ring 20 ms = ~42 ms; RF to phone speaker: ADC ring 20 ms + USB + Android AudioTrack fast path ~20 ms = ~45 ms.

6.5 Security and robustness

- The device never executes anything from the host; all values are range checked (`pmr446_validate`) and the radio is only ever programmed to the 16 legal channels (R1).
- A host cannot disable the time-out, the thermal inhibit or the ERP cap.
- Malformed frames are answered with NAK and counted; counters are exposed on the debug console.

7. Audio path and audio filtering

Half-duplex voice, 300-3000 Hz, FM deviation +/- 2.5 kHz max at 12.5 kHz spacing, plus the data modes that ride on the same path: 1200 Bd AFSK (1200/2200 Hz), FreeDV 2400B (48 kHz modem audio) and, later, 9600 Bd modes. The hardware therefore samples at **48 kHz** with analog corners at 8 kHz; voice is decimated to 16 kHz in firmware (`resample.c`, 96-tap FIR, 0.0 dB at 1-3 kHz, > 50 dB alias rejection). Two audio sources and two sinks exist and are switched by the PTT state, the mode and whether a phone is attached.

```
            RX                                              TX
SA818 AF_OUT -> buffer -> 2nd-order AA LPF -> ADC (16 kHz) -> MRP AUDIO_RX -> phone
             -> speaker amp (standalone)        phone -> MRP AUDIO_TX -> DAC (16 kHz)
                                                  -> reconstruction LPF -> attenuator -> SA818 MIC_IN
local MEMS mic -> preamp -> HPF -> (standalone) -------------------------------^
```

7.1 Receive side

1. **Module AF_OUT**: ~0.3-0.6 Vpp line level, DC biased (**VERIFY**), already de-emphasised and band limited (300 Hz-3 kHz) inside the module, CTCSS tone removed by its 300 Hz high-pass.
2. **Buffer and anti-alias filter**: TLV9062 (rail-to-rail, 3.3 V) unity buffer, then a 2nd-order Sallen-Key low-pass at 8 kHz (Butterworth: R 10 k / 10 k, C 1.4 nF / 2.8 nF), biased at 1.65 V. With 48 kHz sampling the alias band starts at 40 kHz where this filter gives > 28 dB.
3. **ADC**: STM32 ADC1, 12-bit, timer-triggered at exactly 48.000 kHz, DMA double buffer of 960 samples (20 ms). Firmware subtracts the DC bias, scales to s16 and either decimates to 16 kHz (voice to the phone, AFSK demodulator) or feeds FreeDV at 48 kHz. A 1st-order 250 Hz high-pass kills residual CTCSS/hum on the voice path only (data paths stay flat).
4. **Speaker (standalone)**: TPA6211A1 class-AB 3 W mono into an 8 ohm, 28 mm speaker; class-AB rather than class-D so no switching products reach the receiver. Enabled only when no host is attached or the app asks.

7.2 Transmit side

1. **From the phone**: 20 ms PCM frames into a DAC ring buffer (3 frames deep, starts playing when 2 are queued).
2. **DAC**: STM32 DAC1 12-bit at 48 kHz, DMA, output 0.2-3.1 V. Voice from the phone (16 kHz) is interpolated x3 in firmware; AFSK and FreeDV modem audio are generated at 48 kHz directly.
3. **Reconstruction filter**: 2nd-order Sallen-Key 8 kHz (removes the 48 kHz image), then a 1st-order 300 Hz high-pass that can be bypassed by an analog switch (TS5A3159) for flat-path data modes.
4. **Attenuator and limiter**: resistive divider to the module's microphone level (tens of mV; **VERIFY** the module's MIC_IN sensitivity for 2.5 kHz deviation, typically 10-30 mVrms), a soft clipper in firmware before the DAC (peak limiter at -3 dBFS) so that deviation can never exceed the legal limit regardless of what the app sends. Pre-emphasis is done by the module (`AT+SETFILTER` bypass flags all 0).
5. **Local microphone (standalone)**: analog MEMS microphone (Knowles SPU0410LR5H class) with a 2 kHz-cornered noise-gate in firmware, preamp gain 30 dB; muxed to the same reconstruction path through the DAC (the MCU samples the mic on ADC2 and re-synthesises it, which also gives the limiter for free).

7.3 Filters summary

Filter	Where	Corner / type	Purpose
Pre-emphasis 6 dB/oct	module, TX	300 Hz-3 kHz	FM voice SNR, standard PMR
De-emphasis	module, RX		
CTCSS reject HPF	module, RX	300 Hz	removes sub-audible tone

Filter	Where	Corner / type	Purpose
Voice LPF	module, both	3 kHz	deviation / adjacent channel
Anti-alias LPF	hardware, RX	8 kHz 2nd-order	ADC at 48 kHz
Reconstruction LPF	hardware, TX	8 kHz 2nd-order	DAC image
Band HPF	hardware, TX, switchable	300 Hz 1st-order	keep voice above CTCSS band; bypassed for data
Decimation / interpolation FIR	firmware	6 kHz, 96 taps	48 <-> 16 kHz voice path
DC/hum HPF	firmware, RX voice	250 Hz 1st-order IIR	DC bias, residual tone
AFSK band-pass	firmware, RX data	900-2700 Hz biquad	demodulator front end
Peak limiter	firmware, TX	-3 dBFS	deviation limit, R5
Noise gate	firmware, local mic	-45 dBFS	standalone mic hiss
RF bypass	hardware	1 nF + 100 ohm at module pins	446 MHz on audio lines

7.4 Levels

Point	Level
AF_OUT nominal (60 % deviation)	0.4 Vpp (**VERIFY**)
ADC input full scale	3.3 Vpp; nominal signal set to -12 dBFS by buffer gain 2
Wire PCM nominal	-12 dBFS (0.25 full scale) for 60 % deviation; app normalises mic to this
DAC output nominal	1.0 Vpp
MIC_IN nominal	20 mVrms (**VERIFY**), divider 1:30

Ground: audio op-amps and ADC/DAC references on +3V3A, star ground at the MCU analog ground pin, kept away from the buck converter return path.

7.5 Data and digital voice through the same path

Mode	TX source	RX sink	Module filters (`AT+SETFILTER`)
Voice	phone 16 kHz PCM -> x3 -> DAC	ADC -> /3 -> phone	emphasis, HPF, LPF active
AFSK 1200 Bd	`afsk_mod` at 48 kHz -> DAC	ADC -> /3 -> `afsk_demod`	as voice (standard for 1200 Bd)
FreeDV 2400B	`dv_tx_push` modem out 48 kHz -> DAC	ADC 48 kHz -> `dv_rx_push`	emphasis bypassed (flat), HPF/LPF active
Sound-card profile	UAC 48 kHz -> DAC directly	ADC -> UAC 48 kHz	app-selectable, all bypassable
9600 Bd / M17 (future)	48 kHz direct	48 kHz direct	all bypassed; needs the modulator-bandwidth check T-A4

T-A4 (new): measure the module's TX frequency response from MIC_IN with all filters bypassed, 20 Hz-6 kHz; a 9600 Bd mode needs roughly flat to 5 kHz and usable response below 100 Hz. If the module fails, 9600 Bd modes stay out of scope for this module (an alternative RF chip, not a redesign of the audio path, would be needed).

8. Firmware

Code in `firmware/`. Portable modules have no hardware dependency and are unit tested on the host with `make -C firmware test`; the application and the port layer are checked with `make -C firmware check-app`.

8.1 Layout

Path	What
`include/mrp.h`, `src/mrp.c`	MRP framing, CRC, parser with resync, payload pack/unpack
`include/pmr446.h`, `src/pmr446.c`	channel plan, CTCSS table, validation, frequency formatting
`include/sa818.h`, `src/sa818.c`	AT command builder / response parser for the radio module
`include/pd_policy.h`, `src/pd_policy.c`	USB-PD role policy state machine (docs/03)
`include/ptt_guard.h`, `src/ptt_guard.c`	transmit interlocks: keep-alive, TOT, cool-down, inhibit, busy lockout; voice pre-empts data
`bandplan`	PMR446 / HAM 70 cm modes, callsign unlock, frequency and power rules
`ax25`, `afsk`	AX.25 UI frames, HDLC, FCS; Bell 202 modulator and demodulator (16/48 kHz)
`aprs`, `msg`, `beacon`	APRS positions/messages/acks; message retries and dedupe; SmartBeaconing
`arbiter`	who may transmit on the shared antenna path (voice > DV > data > LoRa 433), RF switch position
`resample`	48 <-> 16 kHz FIR
`lora_sx1262`, `lora_aprs`	SX1262 command-level driver (two instances, mock-tested); LoRa-APRS framing
`dv`	FreeDV 2400B glue around the Codec2 library (submodule `third_party/codec2`)
`include/hal.h`	the port API the application needs
`src/app_main.c`	application super-loop and MRP handlers
`port/host/hal_host.c`, `main_host.c`, `freedv_stub.c`	stub port and FreeDV stand-in so the whole application links on Linux (`build/app_host`)
`tests/`	unit tests (`make test`), `vectors.c` reference generator for the Python cross-check
`third_party/codec2`	git submodule, LGPL 2.1; not built on the host by default

Target port (`port/stm32f405/`, not yet written) = CubeMX-generated clock, USB OTG FS, TIM6-triggered ADC1/DAC1 DMA at 16 kHz, USART1 to the module, I2C1 to PD/charger/gauge, plus TinyUSB with a vendor class instance (and CDC / UAC instances optional). Nothing in `src/` touches a register.

8.2 Main loop

```
init: parser, pd_policy, ptt_guard, bandplan, arbiter, msg, beacon, resamplers,
      AFSK demod -> HDLC -> on_ax25_frame, radio_init, two SX1262 (presets), USB profile
loop @ 1 kHz:
  kick watchdog
  USB bulk in -> mrp_parser_feed -> on_frame()
  local PTT button -> ptt_guard_request(LOCAL); sound-card profile: CDC RTS -> ptt_guard_request(HOST)
  voice keyed? -> arbiter_request(FM_VOICE | FM_DV) (pre-empts data / LoRa 433)
  keyed = ptt_guard_tick() -> RF switch to FM, LoRa 433 standby, hal_radio_ptt, audio paths
  data_tx_service(): generate AFSK chunk-wise into the DAC ring; unkey when drained
  audio_rx_service(): ADC 48 kHz -> [DV rx] or [/3 -> AUDIO_RX to phone + AFSK demod]
  lora_service(): SX1262 events (TX done -> release arbiter; RX -> LoRa-APRS -> msg/APRS -> host)
  every 1 s: power status, pd_policy_tick, inhibit, beacon_due -> send, msg_next_tx -> send, STATUS
```

Audio is DMA driven so the loop's jitter does not matter; the loop copies or generates at most a few ms of audio per pass. AFSK is synthesised 64 bits at a time into the DAC ring instead of pre-rendering a whole packet (a 2 s packet at 48 kHz would need 192 KB).

8.3 Interlocks (all in `ptt_guard`, tested)

Interlock	Behaviour
Host keep-alive	PTT from the host is released if no frame arrives within `ttl_ms` (default 2 s). Audio frames count as keep-alive.
Transmit time-out	180 s continuous, then 10 s cool-down in which PTT is refused.
Inhibit	PA/cell temperature > 70 C, SoC < 5 %, module not responding: PTT refused and an active transmission is dropped.
Busy-channel lockout	optional per channel: PTT refused while the squelch is open.
Priority	first keyer wins; the other source can neither steal nor release it.
Power level cap	the H setting is only allowed when the stored antenna gain keeps ERP <= 0.45 W.
Sequencing	radio rail enabled after the MCU drives PTT high (inactive); PTT line has a pull-up so a reset MCU cannot key.

8.3b Operating modes and identity

`bandplan` holds the mode. PMR446 accepts only `SET_CHANNEL`; HAM accepts `SET_FREQ` anywhere on the 12.5 kHz grid between 430 and 440 MHz. HAM is locked until `SET_IDENTITY` stores a syntactically valid callsign (`bandplan_callsign_valid`: 3-7 upper-case alphanumerics, a digit, a letter, ending in a letter). The callsign-SSID becomes the AX.25/APRS/LoRa-APRS source address; in PMR mode the nickname is used instead. Switching back to PMR turns digital voice off and reloads the ISM LoRa preset with the 10 mW ERP power cap.

8.3c Data paths

- **FM data TX** (`data_tx_start`): APRS info -> AX.25 UI (tocall `APZMRP`, path from the beacon settings) -> HDLC with 45 preamble flags (300 ms) -> keyed as `PTT_SRC_DATA` through `ptt_guard` and `arbiter` -> AFSK at 48 kHz into the DAC ring -> unkey when the ring has drained. Voice pre-emption aborts the frame; `msg` retries later.
- **FM data RX**: the 16 kHz voice stream also feeds the AFSK demodulator whenever the squelch is open (or squelch is 0); decoded frames go to the host raw (`DATA_RX`) and through `msg_on_rx` / `aprs_decode` (`MSG_RX`, `APRS_RX`, automatic acks on the same transport).
- **LoRa**: `lora_send()` asks the arbiter (433 shares the antenna, 868 is free), caps 433 power in PMR/ISM mode, configures and transmits; `TX_DONE` releases the arbiter. Receive runs continuously on 868 and on 433 only when the user prefers the LoRa receiver (`LORA_CONFIG.rx_pref`).
- **Digital voice**: `SET_DV` opens FreeDV 2400B (HAM only), bypasses the module's emphasis and the hardware 300 Hz high-pass for a flat path; `AUDIO_TX` frames go through `dv_tx_push` (48 kHz modem audio to the DAC), the ADC stream through `dv_rx_push` (speech back to the phone at 16 kHz).

8.4 Radio module driver

`sa818.c` only builds strings. The sequence after power-up: wait 500 ms, `AT+DMOCONNECT` (expect `+DMOCONNECT:0`), `AT+SETFILTER=0,0,0`, `AT+DMOSETVOLUME=4`, `AT+DMOSETGROUP=0,446.0063,446.0063,0000,4,0000`. Each command is retried 3 times with a 300 ms line timeout. Frequencies are formatted with 4 decimals rounded to 100 Hz; the module's synthesiser grid is finer than the rounding error (the datasheet says "XXX.XXXX"; **VERIFY** that the module accepts `446.0063` and lands on 446.00625 within +/-100 Hz; if a 5-decimal form is accepted, change `pmr446_format_mhz`). RSSI is polled once a second while receiving; raw-to-dBm uses a linear fit (`sa818_rssi_to_dbm`) calibrated at the RF test port.

8.5 Power management

- `pd_policy` decides swaps and source enable; the port turns actions into TPS25751 commands (`SWSr`/`SWDF` 4CC tasks) and BQ25798 register writes.
- Charger: input current limit follows the PD contract; OTG voltage follows the source contract; VSYS min 6.2 V.
- Sleep: with no host and no activity for 5 min the MCU enters STOP mode, wakes on the PTT button, the SQ line, USB attach or the gauge alarm; the radio stays in receive (65 mA) unless the user turns it off with a long press (then only the gauge and PD controller run: < 1 mA).

8.6 Build and test

```
make -C firmware test        # unit tests (gcc)
make -C firmware check-app   # application compiles against hal.h
python3 tools/test_mrp.py    # Python protocol implementation agrees with C
```

Target toolchain: `arm-none-eabi-gcc`, TinyUSB 0.16+, STM32CubeF4 HAL/LL, Codec2 built for Cortex-M4F (the SM1000 build flags are the reference).

Note on the host build of Codec2: `make check-app` links `freedv_stub.c`, a transparent stand-in, so `dv.c` is exercised without the vocoder. A real round trip (`PCM -> FreeDV 2400B -> PCM`) on the host needs the submodule compiled with CMake; that build was not run in the environment where this code was written (executing freshly downloaded third-party build systems was not permitted there), so it is the first thing to do on a developer machine:

```
cmake -S firmware/third_party/codec2 -B /tmp/codec2-build -DCMAKE_BUILD_TYPE=Release -DUNITTEST=OFF
cmake --build /tmp/codec2-build -j --target codec2
# then link build/libmr.a + src/dv.c against /tmp/codec2-build/src/libcodec2.a instead of the stub
```

8.7 CPU budget (to be measured in Phase C)

Task	Estimate on STM32F405 @ 168 MHz
FreeDV 2400B TX or RX (Codec2 1300 + FSK modem)	30-40 % (SM1000 figures)
AFSK demodulator at 16 kHz (4 correlators x 13 taps)	~3 %
Resamplers 48 <-> 16 kHz, 96 taps	~4 %
USB vendor bulk + CDC	~3 %
UAC2 48 kHz isochronous (sound-card profile; DV is off then)	~5 %
Two SX1262 drivers, MRP, control	~2 %

DV and the sound-card profile are never active at the same time, so the worst case is DV + AFSK RX + resampling + USB bulk at roughly 50 %. If the measurement lands above 70 %, the STM32F427 (same pins, 180 MHz, more RAM) or an STM32H7 is the escalation.

9. Mechanical, magnetic attachment and thermal

9.1 Form factor

Item	Value
Body	160 x 76 x 14 mm, PC/ABS shell, TPU bumper edge
Antenna	60 mm stub at the top-left corner, 8 mm boot diameter, flexible TPU over the helix
Mass	~190 g (cells 2 x 65 g, PCB/shell/magnets ~60 g)
Attachment	magnet ring to the phone back (9.2) + 100 mm USB-C to USB-C cable (USB 2.0, 3 A) from the pack's bottom-edge port to the phone's port; cable clips into a groove along the pack edge
Controls	PTT button (side, 12 x 6 mm, tactile under TPU), power/pair button, 2 LEDs (battery, TX/RX), speaker grille, mic port
Ports	USB-C receptacle on the bottom edge, centred, so the cable runs straight down to the phone's port

9.2 Magnetic attachment: Qi2 MPP / MagSafe-compatible ring

The pack glues itself to the phone with the standard magnet ring used by MagSafe and by the Wireless Power Consortium's Qi2 Magnetic Power Profile (MPP). Any phone with the ring built in (iPhone 12 and later, Qi2 Android phones) or any case with a ring snaps onto it with the correct alignment and orientation.

Geometry (follow the published Qi2 MPP / Apple accessory design guideline drawings when ordering; numbers here are the approximate envelope):

Parameter	Value
Ring magnet array	ring of arc-segment NdFeB N52 magnets, outer diameter ~56 mm, inner diameter ~46 mm, thickness 1.0-1.5 mm, alternating-polarity pairs around the ring as in the standard
Orientation magnet	one bar magnet below the ring (the standard's "orientation" feature) so the pack always mounts upright: this keeps the antenna vertical and the cable pointing at the port
Position on the pack	ring centre on the pack's vertical centre line, 80 mm from the top edge (phone coil centres sit near the middle of the phone back; with a 160 mm pack on a 150-165 mm phone this lines up within a few mm)
Hold	> 8 N shear on an iPhone back glass, i.e. it survives a pocket but detaches with a twist; a thin silicone ring on the pack face raises friction
Steel shunt	a 0.3 mm steel backing behind the array (as in commercial rings) roughly doubles the holding force and reduces field inside the pack

Interactions inside the pack:

- The ring is a large conductive loop. Keep the antenna feed and the helix base >= 20 mm from it (the top-left corner placement gives 45 mm) and keep the match network outside its footprint.
- No wireless charging coil in v1 (out of scope), so the ring's interior is free for the cells. If Qi2 charging is added later the coil sits inside the ring and needs its own ferrite; the receiver would then need a Qi coil switching-noise study (110-205 kHz, harmonics negligible at 446 MHz but the coil is a loop antenna for digital noise).
- The phone's compass/magnetometer sees the ring; this is the same as any MagSafe accessory and the phone compensates.
- The speaker's own magnet must sit outside the ring so it does not disturb the ring's polarity pattern.
- Phones without a built-in ring: ship an adhesive ring (standard accessory) or a thin case with one.

Alternative kept in reserve: a snap-on case frame that holds the pack without magnets, for phones that must not have magnets on them (medical devices nearby).

9.3 Stack-up (back of phone -> outside)

```
phone back glass
silicone friction ring / pack face (1.0 mm shell with inner copper foil ground)
magnet ring (1.2 mm) + steel shunt (0.3 mm), potted in the shell face
cells 2 x 60 x 70 x 6 mm, side by side, foam pads
PCB 4-layer 1.2 mm, 150 x 68 mm, components facing outwards
  RF area (top-left 40 x 30 mm) with shield can 4 mm high
  power area (bottom, under the USB-C port) with shield can
  MCU/audio middle
outer shell 1.0 mm with speaker grille and buttons
```

PCB 4-layer: L1 components and RF, L2 solid ground, L3 power, L4 ground / signals. RF CPW on L1 referenced to L2. Thermal vias under the module's PA pad and under the charger.

9.4 Thermal

Source	Dissipation	Condition
SA818 at 1 W RF	~2 W (3.0 W in, 1 W out)	TX, 10 % duty -> 0.2 W average; 180 s continuous allowed
BQ25798 sourcing 18 W	~1.0 W	continuous while charging the phone
TPS62130 at 0.8 A	~0.15 W	TX
LDOs	~0.25 W	
Cells at 2.5 A discharge, 60 mOhm	~0.4 W	sourcing

Worst realistic case: 18 W sourcing plus 10 % TX = ~2.0 W average over a 160 x 76 mm shell = about 6 W/m^2 per side... in practice about 8-10 C above ambient for the shell in free air; stacked on a phone one side is insulated, so expect ~12-15 C rise at the warmest spot, below the 45 C limit (M3) up to 30 C ambient. The 180 s TX time-out limits the module's spike to ~2 W x 180 s = 360 J, heating the RF area about 10 C transiently. Firmware throttles source current at 60 C board temperature and inhibits TX at 70 C PA temperature.

9.5 Ingress and robustness

- IP52 target: gasketed shell seam, membrane over mic and speaker, USB-C open (rated connector), antenna boot bonded.
- Antenna base: brass insert with M3 thread into a boss, PCB soldered to a pin; the boot flexes 90 degrees without transferring load to the solder.
- Drop: 1 m onto concrete, pack alone and on a phone; the magnet ring separates before the phone port or cable takes the load.

10. Regulatory and safety

This product combines a licence-free transmitter with a lithium battery power bank. Both halves carry their own obligations; the radio half dominates the design constraints.

10.1 PMR446 rules (CEPT / EU)

Basis: ECC Decision (15)05 (as amended), ERC Recommendation 70-03 Annex 2, harmonised standard ETSI EN 300 296 (analogue), EN 300 113 (digital, not used).

Rule	Consequence for this design
446.0-446.2 MHz, 16 analogue channels at 12.5 kHz	channel table in `pmr446.c`, firmware can not tune elsewhere
Max 500 mW ERP	antenna gain measured per unit, module power level capped (docs/05)
Integral antenna only	fixed helix, no external connector; the RF test switch is internal
Hand-portable use only, no base stations, repeaters, or fixed installations	the product is a phone accessory; the app does not offer any relay function
Maximum transmit time limited by the equipment (180 s recommended)	`ptt_guard` TOT 180 s
Voice; data allowed in several countries with limits on duty cycle	DATA_TX is a phase-2 feature behind a region flag
No encryption of the voice content	none used; CTCSS/DCS are squelch tones, not privacy
Country variations (some non-EU CEPT members restrict channels or require registration)	region table in firmware; only EU is enabled; the app shows the country list

Outside CEPT the band is not licence-free (in the US 446 MHz is amateur radio). A US variant would be an FRS device under 47 CFR Part 95 (462/467 MHz, 2 W, fixed antenna) and needs separate FCC certification; the module covers the frequency range, so it is a firmware region plus a certification, not a redesign.

10.1b Amateur (70 cm) mode

The HAM mode makes the pack an amateur radio transceiver for 430-440 MHz. Different rules apply to that mode, and the product must keep the two apart:

Point	How the design handles it
Only licensed operators may transmit in the amateur bands	HAM mode is locked until a callsign is stored; the app shows the licence notice; the operator is legally responsible (same as any ham transceiver with a keypad).
Station identification	every AX.25 / APRS / LoRa-APRS frame carries the callsign-SSID; the app reminds the user to identify by voice at the start/end of a QSO and every 10 min (national rules differ).
Equipment built or modified by amateurs is exempt from RED conformity assessment only for the amateur's own use; a product sold to the public is not exempt	the product is CE marked under RED for the PMR446 and ISM functions; the HAM mode is documented as an amateur-radio function of the certified product (as commercial ham transceivers do). EMC and safety assessments cover the device in all modes.
Band plan (IARU Region 1, 70 cm)	432.500 MHz APRS, 433.500 MHz FM calling, 433.775 MHz LoRa-APRS, 438.025 repeater outputs etc.; the app shows the band plan; the firmware only enforces the band edges and the 12.5 kHz grid.
Power	1 W conducted is well under any amateur limit; no ERP cap in HAM mode.
Third-party traffic, encryption	no encryption of content anywhere in the product; messages are plain APRS.

Point	How the design handles it
PMR446 interoperation	a ham may not use the PMR446 channels with more power or an external antenna than PMR446 rules allow: PMR mode keeps its caps regardless of the unlock.

10.1c LoRa under EN 300 220 (short-range devices)

Band	Limit	Firmware enforcement
433.050-434.790 MHz (SRD, Annex B band h1.3 class)	10 mW ERP, no duty limit (h1.3: 10 mW, 100 % duty with <= 25 kHz channel... LoRa 125 kHz falls under the 10 mW / 10 % duty cycle class h1.1-h1.2 in some readings: use **10 mW ERP and 10 % duty** to be safe)	PMR/ISM mode caps 433 LoRa at +10 dBm minus antenna gain, 10 % duty; HAM mode lifts both
869.400-869.650 MHz (band h1.6 class)	500 mW ERP, <= 10 % duty cycle	time-on-air accounting per hour, TX refused above the quota
868.000-868.600 MHz (h1.4)	25 mW ERP, <= 1 % duty	same, if the user tunes there
863-870 MHz general	25 mW ERP, 0.1 % or LBT+AFA	not offered as a preset

The LoRa modules themselves must be CE/RED assessed with the product (their own module certificates help with the radio test report but do not replace the product-level assessment).

10.1d Digital voice and data

- FreeDV 2400B is an open amateur mode (Codec2, LGPL 2.1): no licence issue, HAM mode only.
- Codec2's LGPL requires that users can relink the firmware with a modified library: ship the firmware as object files plus a documented link step, or provide the full source (recommended; the rest of this project is open).
- On PMR446 only analogue FM voice and simple data are offered; the digital PMR446 standards (dPMR446 EN 301 166, DMR446 EN 300 113) are not claimed.

10.2 CE marking (RED 2014/53/EU) and applicable standards

Area	Standard	Notes
Radio spectrum (art. 3.2)	EN 300 296-1/-2 (PMR446 FM), EN 300 220-1/-2 (433 and 868 MHz LoRa as SRD), amateur mode documented	TX power, spurious, adjacent channel power, frequency error, RX sensitivity, blocking, spurious response, with the phone attached as the "host"; LoRa modules tested in the product
EMC (art. 3.1b)	EN 301 489-1, EN 301 489-5	radiated/conducted emissions and immunity, also as a USB-C PD source
Safety (art. 3.1a)	EN IEC 62368-1	audio/video and ICT equipment safety; covers the power bank function, touch temperature, battery enclosure
RF exposure (art. 3.1a)	EN 50566, EN 62209-2	SAR for body-worn/hand-held use at < 20 cm; 0.5 W ERP at 446 MHz
Battery	EN IEC 62133-2	cells and pack
Transport	UN 38.3	mandatory before shipping; 29.6 Wh < 100 Wh (no dangerous goods class for carry-on)
Environmental	RoHS, REACH, WEEE, Batteries Regulation (EU) 2023/1542 (removability requirement from 2027 for portable batteries: design the cell pack to be replaceable with common tools)	

Area	Standard	Notes
USB	USB-IF compliance is voluntary; without certification the USB logos can not be used, but PD interoperability testing is still required by the product (docs/11)	

Conformity route: EN 300 296 is a harmonised standard, so self-declaration with a test report from an accredited lab is possible; many manufacturers still use a Notified Body for the radio part. Budget 4-6 weeks and test samples with the RF test port accessible.

10.3 Battery and power-bank safety requirements folded into the design

- Dual protection (primary IC + secondary OVP with fuse), NTC monitored charging window 0-45 C, discharge window -20 to 60 C.
- Cells rated for the 2 C peak (sourcing 18 W + TX = ~3.5 A peak at 6 V).
- VBUS source: current limit, OVP, short-circuit shutdown, reverse-current blocking; no VBUS unless a sink is detected (Type-C rule, prevents live pins).
- Enclosure: no cell deformation under 1 m drop; vent path for the pouch cells; pouch tabs insulated; no sharp PCB edges facing the cells.
- Labelling: Wh rating, polarity, CE, WEEE, "do not short", manufacturer, model, serial, PMR446 statement.

10.4 User-facing restrictions the app enforces or explains

- Country selector limited to countries where PMR446 is permitted; first run shows the licence-free conditions.
- Speaker-mode PTT only (no "hold at ear" call UI) to keep the RF exposure posture.
- Warning when the phone transmits heavily (hotspot on, poor cellular signal): radio range may drop.

11. Test plan

Bring-up order first, then conformance-style tests. Test IDs refer to the requirement they cover.

11.1 Bring-up order

1. **Power only, no cells**: feed 7.4 V from a bench supply with 1 A limit on the cell pads; check rails (+4V0, +3V3D, +3V3A), ripple, sequencing. Then attach cells through the protector; verify charge/discharge FET control.
2. **PD controller**: program TPS25751 configuration, attach a PD analyser (e.g. a Type-C sniffer such as a Twinkie/PD analyser) and a laptop: observe contracts, Try.SRC behaviour, swaps.
3. **MCU and USB**: enumerate the vendor interface on Linux, run `python3 tools/mrp.py --usb info`.
4. **Radio module**: `AT+DMOCONNECT` over the debug console pass-through, then `tools/mrp.py channel 1`; confirm RX on a service monitor through the RF test port.
5. **Audio**: RX tone from the monitor to the phone/PC; TX tone from the PC through the module into the monitor; set levels (docs/07).
6. **Filters**: VNA sweep of the LPF and match through the test port and a temporary U.FL on the antenna side before the antenna is fitted.
7. **Antenna**: tune the helix on the assembled pack on a phone; measure ERP on an OATS or in a chamber by substitution.
8. **Phone**: Android app on three phones (recent Pixel, Samsung, one mid-range), iOS with the UAC profile.

11.2 Radio tests (conducted at the RF test port unless stated)

ID	Test	Method	Limit / expectation
T-R1	Channel frequencies	counter on each of the 16 channels	+/- 1.5 kHz (EN 300 296 for 12.5 kHz), expect < 300 Hz
T-R2a	Conducted power H/L	power meter, all channels, 6.0/7.4/8.4 V battery, -10/25/55 C	per module spec; H ~ +30 dBm, L ~ +27 dBm
T-R2b	ERP	substitution on OATS, pack on phone, 3 orientations	<= 27 dBm ERP with the selected level; record antenna gain
T-R5	Spurious emissions TX	spectrum analyser 9 kHz-4 GHz with notch at the carrier	<= -36 dBm < 1 GHz, <= -30 dBm > 1 GHz; expect 2nd harmonic < -60 dBm
T-R5b	Adjacent channel power	ACP at +/-12.5 kHz with 1 kHz tone at 60 % and max deviation	<= -60 dBc or 0.2 uW
T-R5c	Max deviation	1 kHz tone at +20 dB over nominal input	<= +/- 2.5 kHz (limiter proof)
T-R6	Receiver sensitivity	12 dB SINAD at the port, each channel	<= -116 dBm; also with the phone attached, screen on, data transfer running
T-R6b	Self-generated spurs	sweep channels, listen with squelch 0 while USB streaming, charging, sourcing	no birdie above -120 dBm equivalent on any channel
T-R7	Desense by the phone	phone forced to max cellular TX (test mode or weak-signal box) on B20/B28/B8/B3/n78; measure sensitivity	<= 6 dB degradation with option A; decide option B
T-R8	Pack TX into phone	pack at 1 W while the phone runs a GNSS fix and a data throughput test	no GNSS loss, < 10 % throughput drop
T-R9	Blocking / spurious response	EN 300 296 methods	per standard
T-R10	Busy-channel lockout, TOT, keep-alive	functional with a second radio	180 s TOT, 2 s keep-alive, lockout refuses PTT

11.2b Data, LoRa, digital voice and modes

ID	Test	Expectation
T-D1	AFSK packet into Direwolf / a hardware TNC from the pack's audio (and from `tools/afsk_wav.py`)	100 % decode at good signal; frame bytes identical to `firmware/build/vectors`
T-D2	APRS beacon received by an iGate (HAM mode, 432.500 MHz FM and 433.775 MHz LoRa)	position appears on aprs.fi with the callsign-SSID
T-D3	Two packs exchange messages over FM and over LoRa 433/868; acks and retries	delivery and `MSG_STATUS` acked; duplicate suppression
T-D4	Voice pre-emption	PTT during a data frame keys voice within 50 ms; data retried later
T-D5	Busy-channel behaviour for data	no data TX while the squelch is open; back-off
T-A3	AFSK deviation	1200/2200 Hz tones at 3.0 kHz +/- 0.3 kHz peak deviation
T-A4	Module modulator flatness, all filters bypassed, 20 Hz-6 kHz	record; decides 9600 Bd / M17 feasibility
T-A5	FreeDV 2400B over the air to an SM1000 or a PC running FreeDV	sync and intelligible audio at >= -110 dBm
T-L1	LoRa 433/868 TX power and spurious (EN 300 220) at both module ports	<= limits; 433 PMR mode <= 10 mW ERP
T-L2	FM 1 W TX leakage into the LoRa 433 input	<= +5 dBm measured at the module pin; module survives 1000 keyings; limiter stuffed if > +5 dBm
T-L3	LoRa 868 TX while receiving weak FM	no audible click / desense <= 3 dB
T-L4	868 duty-cycle accounting	TX refused after 6 min of air time per hour in 869.4-869.65
T-M1b	Mode lock	HAM mode refused without a callsign; PMR caps unchanged after unlock
T-U5	Sound-card profile with Direwolf (RTS PTT) and FreeDV GUI on a PC, and as a headset on iOS	works without any driver or app of ours

11.3 Power and USB-C tests

ID	Test	Expectation
T-P2	Source PDOs with an e-load through a PD tester: 5 V/3 A, 9 V/2 A for 1 h	voltage within PD limits, shell < 45 C
T-P3	Sink from 5 V/1.5 A legacy, 9 V/3 A and 15 V/2 A PD chargers	2 A charge current, correct termination, timer
T-P4	Radio rail transient when keying at 6.0 V battery	+4V0 dip < 100 mV, no module reset, no audio click > -40 dBFS
T-P4b	Radio rail ripple in RX	< 10 mVpp at the module, no 2.5 MHz products in the audio
T-P5	Reserve behaviour	drain to 20 %: source stops, PR_Swap requested, pack trickle-sinks 100 mA; charge to 30 %: source resumes
T-P8	Faults	VBUS short while sourcing; 20 V applied while sourcing; CC shorted to VBUS; reverse current
T-P9	Protector	OV, UV, OC, short, over-temperature trip points
T-PD1	Role negotiation matrix	6 phones x {Try.SRC win, Try.SNK win}, 3 chargers, 2 PCs, legacy A-to-C
T-PD2	Phone user enables reverse charging	pack rejects while sourcing, accepts when below reserve
T-PD3	Hot plug 100 cycles	no stuck state; watchdog never fires

11.4 Interface, audio and app tests

ID	Test	Expectation
T-U1	Enumeration on Android 12-15, Linux, Windows 11 (WinUSB), macOS	vendor interface claimable without driver install
T-U2	End-to-end latency	mic tone edge at the phone to RF demodulated edge: < 60 ms; reverse likewise
T-U2b	Jitter / loss	1 h stream, count `seq` gaps
T-U3	UAC profile	iOS sees a headset, voice both ways, pack PTT works
T-U4	STATUS content	compare with bench meters: battery +/- 50 mV, current +/- 100 mA, SoC +/- 5 %
T-A1	Audio frequency response TX and RX	300 Hz-3 kHz within +1/-3 dB after emphasis
T-A2	Standalone mode	PTT button, speaker, mic with no phone; CTCSS decode with a consumer PMR446 radio

11.5 Mechanical, thermal, environmental

ID	Test	Expectation
T-M1	Magnet hold	shear and pull force on an iPhone back and on a Qi2 case; 1000 attach cycles
T-M2	Antenna boot flex	10 000 cycles of 90 degree bend; VSWR change < 0.3
T-M3	Thermal map at 18 W + 10 % TX at 25 C and 35 C ambient	surface < 45 C / < 50 C
T-M4	Drop 1 m, 6 faces, on a phone	no cell damage, RF performance unchanged
T-M5	IP52	per IEC 60529
T-E1	EMC pre-scan radiated emissions 30 MHz-6 GHz while sourcing and receiving	EN 301 489-1 class B margin > 6 dB
T-E2	ESD 8 kV contact on antenna boot, USB shell, buttons	no reset of the radio path, no damage

12. Packet data, APRS and text messages over FM

Everything in this chapter rides on the ordinary FM voice path of the SA818: audio tones into MIC_IN, audio tones out of AF_OUT. No hardware beyond the voice design is required; the firmware modules are `afsk`, `ax25`, `aprs`, `msg`, `beacon` and `arbiter` (all host-tested).

12.1 Physical layer: Bell 202 AFSK, 1200 Bd

Item	Value
Tones	mark 1200 Hz, space 2200 Hz, continuous phase
Rate	1200 Bd, NRZI (a 0 bit toggles the tone, a 1 bit keeps it)
Deviation	set the DAC level so that the tones give ~3 kHz peak deviation (T-A3); the module's pre-emphasis stays on, as with every handheld TNC set-up
TX sequencing	key PA, 300 ms of flags (TX delay, 45 flags), frame, 3 flags, unkey. TX delay is configurable 100-500 ms
RX	48 kHz ADC -> `resample` to 16 kHz -> `afsk_demod` (band-pass, one-bit quadrature correlators at both tones, PLL bit sync) -> `hdlc_rx` (NRZI, de-stuffing, FCS)

Simulated performance of the demodulator (`tests/test_afsk.c`, white noise over the whole 16 kHz band, 100 packets of 90 bytes):

SNR in 16 kHz band	equivalent in a 3 kHz voice band	packets decoded
20 dB	24 dB	100 %
10 dB	14 dB	100 %
6 dB	10 dB	100 %
4 dB	8 dB	98 %
2 dB	6 dB	63 %
0 dB	4 dB	1 %

It also tolerates +/-2 % sample-rate error and 35 dB of level variation. Real FM channels add distortion and fading; expect packets to decode down to about the 12 dB SINAD point of the receiver, which is what hardware TNCs achieve.

12.2 Link layer: AX.25 UI frames

Standard AX.25 v2 unnumbered information frames: destination, source, up to 8 digipeaters, control 0x03, PID 0xF0, information field, FCS CRC-16/X-25, HDLC flags and bit stuffing. Destination ("tocall") is `APZMRP` (experimental prefix APZ; register an APxxxx tocall before release). Default path `WIDE1-1` (one hop) to be polite on the VHF/UHF APRS network; the app can set `WIDE1-1,WIDE2-1` or none.

12.3 Application layer: APRS

Function	Format	Where
Position	`!DDMM.hhN/DDDMM.hhE[CSE/SPD/A=aaaaaa comment`, symbol table/code configurable (default `/[` jogger, `/>` car when speed > 20 km/h)	`aprs_encode_position`
Message	`:ADDRESSEE:text{id}` (id present = ack requested)	`aprs_encode_message`
Ack / reject	`:ADDRESSEE:ackID` / `rejID`	`aprs_encode_ack`
Decoder	uncompressed positions with/without timestamp, course/speed, messages, acks, status; compressed and Mic-E positions are reported as unknown (phase 2)	`aprs_decode`

Position source: the phone sends `POSITION` frames (lat/lon/alt/speed/course) from its GNSS over MRP; the pack has no GNSS of its own. In standalone use the last known position is beaconed with an age flag in the comment.

12.4 Text messages (`msg.c`)

- Outbox of 8, each message gets an id 1..99999, sent with `{id}`.
- Retries after 10, 20 and 40 s; after 3 attempts the message is marked failed and the app is told (`MSG_STATUS`).
- Incoming messages addressed to our identity (callsign-SSID in HAM mode, nickname in PMR mode) are delivered to the app (`MSG_RX`) and acknowledged; duplicates (same sender and id within the last 16) are acknowledged again but flagged so the app does not show them twice.
- Messages to a group use addressee `ALL` or a group nickname and no id.
- The same layer is used over LoRa (docs/13); the `transport` field of `SEND_MSG` selects FM, LoRa 433 or LoRa 868.

12.5 Beacons (`beacon.c`)

Fixed interval (default 10 min) or SmartBeaconing: 30 min when stationary, 60 s at >= 90 km/h, interpolated in between, plus a beacon on a course change larger than 28 deg + 255/speed after at least 15 s. A hard floor of 30 s between rate beacons protects the channel.

12.6 Channel arbitration (`arbiter.c`, `ptt_guard.c`)

- Voice (phone or button) > digital voice > FM data > LoRa 433 on the shared antenna path. Voice pre-empts an ongoing data frame immediately (the frame is lost and retried later); data never interrupts voice and is refused while voice is keyed.
- Data TX uses the same `ptt_guard` as voice (source `PTT_SRC_DATA`), so the time-out, thermal inhibit and busy-channel lockout apply. Data frames use busy-channel detection by default (wait up to 5 s for the squelch to close, then random back-off 0.5-2 s).
- Duty cycle: beacons and messages are small (< 1 s of TX); the firmware limits data to 10 % averaged over 60 s in PMR mode (country rules vary; docs/10).

12.7 PMR446 and data

ECC/DEC/(15)05 allows data on PMR446 in general, but several administrations apply conditions (no store-and-forward, duty cycle, no fixed use). The pack's data features in PMR mode are therefore: text messages and position exchange between packs, with the nickname as identity and the 10 % duty limit. APRS beacons into the amateur network are HAM mode only.

12.8 Bench tools

- `tools/afsk_wav.py` writes an AX.25/AFSK packet as a 48 kHz WAV so it can be played into a real TNC or Direwolf for an independent decode check, and decoded by the C demodulator test.
- `tools/mrp.py msg`, `beacon`, `pos`, `identity` drive the features from a PC.

13. LoRa: 433 MHz and 868 MHz

13.1 Why two modules

There is no production dual-band LoRa module: the SX1262 chip tunes 150-960 MHz, but every module's matching, harmonic filter and PA are built for one band. Two 22 dBm single-band modules (E22-400M22S and E22-900M22S class, SX1262 inside, TCXO, ~6 EUR each) are simpler and lower risk than a custom dual-band front end on a first board.

	LoRa 433	**LoRa 868**
Module	E22-400M22S class (SX1262 + TCXO, +22 dBm)	E22-900M22S class
Antenna	shares the 439 MHz helix through an SPDT switch (PE4259)	own printed inverted-F at the opposite top corner (docs/05)
Use	LoRa-APRS 433.775 MHz, pack-to-pack 433.900 MHz, ham experiments, Meshtastic EU_433 PHY	pack-to-pack and Meshtastic EU_868 PHY at 869.525 MHz, ISM 500 mW band
Legal power	PMR/ISM mode: 10 mW ERP (EN 300 220, 433.05-434.79 MHz); HAM mode: +22 dBm	869.4-869.65 MHz: 500 mW ERP, 10 % duty; 868.0-868.6: 25 mW, 1 %
Concurrency	alternates with FM (voice has priority); listens only when the user prefers the LoRa receiver over FM	independent of FM; RX continuous

13.2 Antenna sharing on 433 MHz

```
SA818 -> LPF ----\
                  SPDT (PE4259) -> BPF 439/50 -> test switch -> match -> helix
LoRa433 module --/
```

- The switch is controlled by the arbiter (`arbiter_antenna()`): FM during FM TX and by default in RX; LoRa 433 during its TX and when the user selects `rx_pref = LORA433`.
- Isolation: PE4259 gives ~25 dB at 450 MHz. With 1 W FM TX the LoRa module would see about +5 dBm, inside the SX1262 absolute maximum (+10 dBm) but not comfortable. The firmware holds the LoRa 433 module in standby during FM TX, and the PCB has a footprint for a shunt PIN diode limiter (BAP64-02 class) on the LoRa branch. Measured in T-L2.
- The BPF (439 MHz, 50 MHz wide) is in the common path, so LoRa-APRS at 433.775 MHz passes with ~1.2 dB loss and the LoRa receiver gets the same protection from the phone's cellular transmitter as the FM receiver.
- Consequence to accept: continuous LoRa-APRS or mesh *reception* on 433 MHz mutes FM reception. If that matters more than board space, a second 433 antenna (another printed IFA) removes the restriction; the schematic keeps the option as a stuffing variant.

13.3 Radio settings (`lora_sx1262.c` presets)

Preset	**Frequency**	**SF/BW/CR**	**Sync**	**Preamble**	**Power**	**Time on air, 50 B**
LoRa-APRS (EU)	433.775 MHz	SF12 / 125 kHz / 4:5	0x12	8	22 dBm (HAM)	2.3 s
Pack-to-pack 433	433.900 MHz	SF9 / 125 kHz / 4:5	0x12	8	10 dBm (ISM)	0.33 s
Pack-to-pack 868	869.525 MHz	SF9 / 125 kHz / 4:5	0x12	8	22 dBm	0.33 s
Meshtastic EU_868 LongFast (PHY only)	869.525 MHz	SF11 / 250 kHz / 4:5	0x2B	16	22 dBm	1.1 s

The driver sets LDRO automatically when the symbol time exceeds 16 ms, uses the SX1262 +22 dBm PA table, applies the TX-clamp errata and 140 mA OCP, and reports RSSI/SNR per packet. Image calibration

bands are chosen from the frequency.

13.4 Frame formats

- **LoRa-APRS**: `'<' 0xFF 0x01` + `SRC>DST,PATH:INFO` (TNC2 text). INFO is ordinary APRS, so the same `aprs.c`/`msg.c` code produces positions, messages and acks; iGates forward them to the APRS-IS. (`lora_aprs.c`)
- **Pack-to-pack**: the same LoRa-APRS framing on the pack frequencies, with nicknames in PMR mode. One code path, two bands.
- **Meshtastic**: the PHY preset lets the module hear and be heard on a Meshtastic channel at the radio level, but the Meshtastic packet format (protobuf header, AES channel key, flooding/routing, node DB) is not implemented. Doing that properly is a project of its own; it is listed as a later phase so nobody expects mesh interoperability from v1.

13.5 Duty cycle and power enforcement

- 868 MHz: the firmware accumulates time on air per hour per sub-band and refuses TX beyond 10 % (869.4-869.65) or 1 % (868.0-868.6).
- 433 MHz in PMR/ISM mode: power capped at +10 dBm conducted (0 dBd antenna assumption; adjust from the measured helix gain) to stay under 10 mW ERP; HAM mode lifts the cap to +22 dBm.
- Both LoRa transmitters are inhibited by the same thermal and battery interlocks as the FM PA.

13.6 Hardware notes for the schematic

- Each module: 3.3 V (+3V3R rail, 150 mA TX each), SPI (shared bus, separate NSS), BUSY, DIO1 (IRQ), NRST, TXEN/RXEN (driven by the MCU via the driver's `rf_switch` callback; DIO2 switch mode is also supported).
- Keep the 868 IFA >= 30 mm from the helix base and >= 20 mm from the magnet ring; a 10 pF series capacitor pad and a shunt inductor pad for tuning.
- 433 branch: 50 ohm CPW to the SPDT; optional shunt PIN limiter footprint.
- Shield both modules (they come canned) and route their SPI away from the audio front end.

14. Digital voice: FreeDV 2400B

14.1 Why this mode

Candidate	Vocoder	Modem / path	Verdict
FreeDV 2400B	Codec2 1300 bit/s	2-FSK-class modem at 48 kHz designed to pass through the mic/speaker path of an ordinary FM radio	**Chosen.** Open (LGPL 2.1), proven on the STM32F405 (the SM1000 is exactly that chip), no hardware change. Interoperates with other FreeDV 2400B stations.
M17	Codec2 3200	4FSK 4800 sym/s needs a flat modulator path to DC and ~5 kHz	Only if T-A4 shows the SA818's bypassed path is flat enough; otherwise it needs a different RF chip (CC1200 class). Kept as a software option.
DMR / dPMR446	AMBE+2 (licensed)	4FSK, standard-defined	Excluded: vocoder licensing and type approval for digital PMR446.
Codec2 over LoRa	Codec2 700C/1300	LoRa SF7 BW250 (~11 kbit/s)	Possible pack-to-pack novelty, not interoperable; not planned.

14.2 How it is wired

```
phone mic 16 kHz --MRP AUDIO_TX--> dv_tx_push: 16->8 kHz, Codec2 encode, modem -> 48 kHz PCM -> DAC ->
SA818 MIC_IN (flat path)
SA818 AF_OUT -> ADC 48 kHz -> dv_rx_push: modem demod (freedv_nin() sized blocks), Codec2 decode, 8->16
kHz --MRP AUDIO_RX--> phone
```

- `SET_DV 1` (HAM mode only) opens the codec, bypasses the module's pre/de-emphasis (`AT+SETFILTER=1,0,0`) and the hardware 300 Hz high-pass (`hal_audio_flat_path`), and routes audio through `dv.c` instead of the resamplers. `SET_DV 0` or leaving HAM mode restores analogue voice.
- Frame timing: 40 ms speech frames (320 samples at 8 kHz) produce 1920 modem samples at 48 kHz; the receiver consumes `freedv_nin()` samples per call (the modem adjusts it for clock offset). `dv.c` buffers accordingly and is unit tested with a transparent stand-in for the library.
- PTT, time-out, arbitration and interlocks are identical to analogue voice; the arbiter kind is `TX_FM_DV` (same priority as voice).
- The STATUS frame reports DV enabled and sync; the pack LED blinks the sync state so standalone use (button PTT) also works in DV.

14.3 Resource and licence notes

- CPU: Codec2 1300 + the 2400B modem take about a third of an STM32F405; measured on the SM1000. Budget in docs/08.
- Flash/RAM: Codec2 with only the 1300 mode and the 2400B modem compiled in (`CODEC2_MODE_EN_DEFAULT=0`, enable 1300 and 2400B) is well under 200 KB flash and 40 KB RAM.
- Licence: LGPL 2.1 (`firmware/third_party/codec2/COPYING`). Publish the firmware source or at least the object files and link instructions.
- Audio levels: FreeDV expects the FM path to be roughly flat 300-2800 Hz with the 2400B tones; set the DAC drive for ~2.5 kHz deviation (T-A5).

14.4 What stays open

- Real round-trip test of `dv.c` against the compiled library (docs/08: not run in the environment this was written in).
- Listening tests on the real FM path: deviation setting, squelch tail behaviour (open squelch while DV is enabled, the modem's own sync decides).
- M17 feasibility measurement (T-A4).

15. ESP32-C5 option: a third radio family and a wireless phone link

Suggestion under review: use the ESP32-C5, which can drive its own radios as a raw transceiver on 2.4 GHz and 5 GHz (raw 802.11 frame injection and promiscuous receive, the Espressif 802.11 LR long-range PHY, BLE 5 coded PHY, IEEE 802.15.4). This chapter evaluates where it fits. Numbers for the C5 are from the Espressif datasheet and SDK documentation and should be re-checked against the current revision (**VERIFY**).

15.1 What the ESP32-C5 is

Item	ESP32-C5 (WROOM-1 module)
CPU	single-core RISC-V, 240 MHz, 384 KB SRAM (+ optional PSRAM), low-power core
Radios	Wi-Fi 6 dual band 2.4 / 5 GHz (20 MHz), Bluetooth 5 LE (incl. coded long-range PHY), IEEE 802.15.4 (Zigbee / Thread)
Raw radio access	`esp_wifi_80211_tx` raw frame injection, promiscuous RX, ESP-NOW (connectionless 802.11 data frames), 802.11 LR proprietary PHY (2.4 GHz only, 250 kbit/s class, ~ -105 dBm sensitivity, about 1 km line of sight), BLE advertising/GATT, 802.15.4 raw frames
TX power	up to about +20 dBm at 2.4 GHz, less at 5 GHz; the module is pre-certified (CE/FCC) with its PCB antenna
Peripherals	USB Serial/JTAG (a fixed CDC device, **not** a USB OTG controller), 12-bit ADC, no DAC, I2S, SPI, UART, SDIO slave, TWAI
Supply	3.3 V, 100-350 mA bursts during Wi-Fi TX, < 100 uA deep sleep

What it cannot do: anything below 2.4 GHz. PMR446, the 70 cm band and the 433/868 MHz LoRa bands stay with the SA818 and the SX1262s. The published hacks that make ESP32 chips radiate at arbitrary sub-GHz frequencies through clock-output harmonics are not usable in a product (broadband spurious emissions, no filtering, illegal).

15.2 Three ways to use it

Option A: ESP32-C5 as the main MCU (replaces the STM32F405)

For	Against
One chip does control, audio DSP, Wi-Fi/BLE to the phone, pack-to-pack 2.4/5 GHz	**No USB OTG device**: it cannot present the vendor MRP interface, the UAC sound-card profile or a CDC-with-RTS PTT. The whole USB phone link and the Digirig compatibility would be lost; the phone link would have to be BLE/Wi-Fi only
Cheaper than STM32F405 + nothing else	No DAC (external I2S codec needed), single core shared with the Wi-Fi/BLE stacks (Codec2 + modem + stacks is tight), radio firmware is closed binary
Pre-certified module simplifies the radio part of RED for 2.4/5 GHz	A Wi-Fi transmitter in the pack means another coexistence study next to a -120 dBm receiver (out of band, so manageable, but it adds test time)

Verdict: not for this product as specified. It would be the right core for a *different* product, a wireless-only accessory with USB used for power alone.

Option B: ESP32-C5 as a connectivity co-processor next to the STM32F405 (recommended)

The STM32 keeps USB, audio, the UHF radios and the interlocks. The C5 sits on a UART (2 Mbit/s, with flow control) or SDIO and offers:

Function	How	What it buys
Wireless phone link	MRP frames over BLE GATT (control, status, messages) and over Wi-Fi UDP (PCM audio, MRP framing unchanged) in SoftAP or station mode	The phone no longer needs the cable for data: the pack can sit in a bag or on a window while the phone is in hand. **iOS gets full app support** without MFi, because BLE and Wi-Fi are open to iOS apps while USB vendor interfaces are not
Pack-to-pack 2.4 GHz long range	ESP-NOW over the 802.11 LR PHY, raw frames with our own header; MRP messages, positions, even voice (Codec2 1300 needs 1.3 kbit/s)	~1 km line of sight, licence-free, higher throughput than LoRa, no duty-cycle rule beyond EN 300 328
Pack-to-pack 5 GHz	ESP-NOW / raw frames at 5 GHz	Higher data rate for short range (files, images). In the EU outdoor use is confined to 5.470-5.725 GHz with DFS/TPC or 5.725-5.875 GHz at 25 mW SRD; 5.15-5.25 GHz is indoor only. Useful, but 2.4 GHz LR is the long-range tool
BLE coded PHY	advertising beacons with position	A low-power "where is my group" beacon for other phones even without our app running
802.15.4 / Thread	optional	sensor or home-automation tie-in; not a goal here
Over-the-air update	Wi-Fi	firmware updates for both MCUs without a PC

Costs: one ESP32-C5-WROOM-1 module (about 3-4 EUR), its own 3.3 V budget (peak 350 mA, shares +3V3R with the LoRa modules only if the regulator grows to 1 A), a dual-band 2.4/5 GHz PCB or chip antenna with keep-out at a third corner (or the module's built-in antenna, which keeps its certification), one UART, one GPIO for enable/boot, 20 x 25 mm of board, and the coexistence tests T-C1..T-C3 below.

Option C: later add-on

Keep the footprint and the UART on the board, populate the module in a second build. Costs almost nothing now and keeps the decision open. This is the minimum I would do.

15.3 Recommendation

Option B, in two steps: lay out the board for the module (Option C) and bring up the wireless phone link first, because it changes the user experience most (no cable for data, iOS support). Pack-to-pack 2.4 GHz LR comes second; 5 GHz raw frames last, as a short-range extra.

The protocol work is small because MRP is transport agnostic: the same frames travel over USB bulk, BLE (a GATT characteristic carrying frames with notification), Wi-Fi UDP (one or more frames per datagram) and the STM32-to-C5 UART. The C5 firmware is a bridge plus the ESP-NOW/LR pack-to-pack layer; it does not touch the radios' safety interlocks, which stay in the STM32.

15.4 Design impact if Option B is adopted

Area	Change
Architecture (docs/02)	new block "ESP32-C5 connectivity", UART2 2 Mbit/s with RTS/CTS, EN and BOOT GPIOs, +3V3W rail (1 A)
Power (docs/02, 03)	+0.1 W average with BLE on and Wi-Fi off; +0.5-1 W while streaming audio over Wi-Fi; deep sleep when the cable is attached (USB link preferred)
RF (docs/04)	third antenna at the bottom-left corner (the magnet ring and the USB port constrain placement); 2.4 GHz TX is 5.4 x 446 MHz: no harmonic relation; the FM LPF/BPF give > 70 dB; the LoRa 868 front end needs the module's own 2.4 GHz rejection (SX1262 input filter) checked in T-C2
Protocol (docs/06)	MRP over BLE: service UUID, one write characteristic (host to pack) and one notify characteristic (pack to host), 244-byte MTU fragments; MRP over UDP: port, discovery via mDNS; same message set
Firmware (docs/08)	`link_mux`: the STM32 accepts MRP from USB, UART-from-C5 (BLE/Wi-Fi) with a single active "voice host" at a time; C5 firmware in `firmware/esp32c5/` (ESP-IDF)

Area	Change
Regulatory (docs/10)	EN 300 328 (2.4 GHz), EN 301 893 (5 GHz), covered mostly by the module's certification; product-level EMC still required
Test plan (docs/11)	T-C1 Wi-Fi TX vs FM/LoRa receiver desense; T-C2 LoRa 868 blocking by 2.4 GHz; T-C3 BLE audio-control latency; T-C4 ESP-NOW LR range on the pack antenna
Mechanical (docs/09)	module placement away from the magnet ring and the hand

15.5 Open questions for review

1. Is the wireless phone link (BLE + Wi-Fi) wanted for v1, or is USB-only acceptable for the first board (Option C now, B later)?
2. Pack-to-pack at 2.4 GHz LR vs LoRa 868: both? LR is faster and licence friendly; LoRa 868 reaches further through walls. If one has to go, which?
3. Is 5 GHz worth the antenna complexity given the EU outdoor restrictions?
4. Should the C5 own the BLE "find my group" beacon even when the pack is otherwise asleep?

Appendix A. Bill of materials

Source: `hardware/bom.csv`.

ref	qty	part	value_or_type	package	function	alternative	notes
U1	1	STM32F405RGT6	ARM Cortex-M4 168 MHz USB FS 2xDAC	LQFP-64	MCU	STM32F407VGT6	TinyUSB vendor + CDC + UAC
U2	1	TPS25751D	USB-C PD3.0 DRP controller I2C	VQFN-32	USB-PD	FUSB302B + MCU PD stack	Try.SRC; EEPROM/I2C configured
U3	1	BQ25798	1-4S buck-boost charger with OTG	QFN-29	charger / power path	BQ25792	2S 8.4 V 2 A charge; OTG 5/9/12 V
U4	1	BQ34Z100-G1	multi-cell fuel gauge I2C	TSSOP-14	gauge	firmware coulomb count on BQ25798 ADC	5 mOhm sense
U5	1	S-8252AAx (Seiko)	2S Li-ion protector	SOT-23-6	primary protection	Nisshinbo R5478	with dual N-FET Q1
U6	1	BQ7718	secondary over-voltage protector	SOT-23	secondary protection		drives chemical fuse F1
U7	1	TPS62130	3 A sync buck adjustable 2.5 MHz	QFN-16	+4V0 radio rail	TPS62133 / TPS62913	forced PWM; LC post-filter
U8	1	TPS7A2033	300 mA low-noise LDO 3.3 V	SOT-23-5	+3V3D		from +4V0
U9	1	LP5907-3.3	250 mA ultra-low-noise LDO	SOT-23-5	+3V3A analog	TPS7A2033	separate star ground
U10	1	SA818U-V4 (NiceRF)	UHF FM transceiver module 400-480 MHz 1 W	module 36x19 mm	FM radio (PMR446 + 70 cm)	DRA818U (Dorji)	shielded; UART 9600
U14	1	E22-400M22S (Ebyte)	SX1262 LoRa module 410-493 MHz +22 dBm TCXO	module 14x20 mm	LoRa 433 (shares helix)	NiceRF LoRa1262-433	SPI; TXEN/RXEN; BUSY; DIO1
U15	1	E22-900M22S (Ebyte)	SX1262 LoRa module 850-930 MHz +22 dBm TCXO	module 14x20 mm	LoRa 868 (own antenna)	NiceRF LoRa1262-868	SPI; TXEN/RXEN; BUSY; DIO1
U16	1	PE4259	SPDT RF switch +34 dBm	SC-70-6	helix: FM chain / LoRa 433	Skyworks SKY13351	arbiter-driven; ~25 dB isolation
U17	1	TPS7A2033	300 mA low-noise LDO 3.3 V	SOT-23-5	+3V3R LoRa modules		separate rail keeps LoRa SPI noise off the audio LDO
U18	1	TS5A3159	analog SPDT switch	SC-70-6	bypass of the 300 Hz TX high-pass (flat path for data/DV)		

ref	qty	part	value_or_type	package	function	alternative	notes
D4	1	BAP64-02 (optional)	shunt PIN diode limiter	SOD-523	LoRa 433 input protection during FM TX		stuff if T-L2 > +5 dBm
ANT2	1	printed meander IFA 40x8 mm keep-out			868 MHz LoRa antenna		series 10 pF + shunt L tuning pads; >= 30 mm from the helix
U11	1	TLV9062	dual RRIO op-amp	SOT-23-8 / VSSOP-8	audio buffer / AA filter	MCP6002	
U12	1	TPA6211A1	3 W class-AB mono speaker amp	MSOP-8	standalone speaker		class-AB to avoid switching noise
U13	1	TLV9062	dual op-amp	VSSOP-8	DAC reconstruction + mic preamp		
MK1	1	SPU0410LR5H-QB	analog MEMS microphone	3.76x2.95 mm	standalone mic	electret 4 mm	
LS1	1	8 ohm 1 W 28 mm speaker			standalone speaker		magnet outside the ring area
J1	1	USB4105-GF-A (GCT)	USB-C 16-pin receptacle mid-mount	SMD	USB-C	Amphenol 12401610E4#2A	power + USB2 + CC only
J2	1	MM8130-2600 (Murata)	RF test switch connector	SMD	RF test port	MM8430	internal; not user accessible
D1	1	TPD2E2U06 or USBLC6-2SC6	USB D+/D- TVS 0.5 pF	SOT-23-6 / DFN	ESD USB		
D2	1	PESD0402-140	RF ESD diode 0.1-0.2 pF	0402	antenna ESD		with shunt L of the match
D3	1	SMBJ24A	VBUS TVS	SMB	USB VBUS		
L1	1	2.2 uH 8 A shielded		7x7 mm	BQ25798 inductor		
L2	1	1.0 uH 3 A shielded		4x4 mm	TPS62130		
L3	1	1 uH + FB 600R@100MHz		0805	+4V0 post-filter		
L_LPF2,L_LPF6	2	22 nH Q>=60 wirewound		0603	TX LPF	Coilcraft 0603HP-22N / Murata LQW18AN22NG	
L_LPF4	1	24 nH Q>=60 wirewound		0603	TX LPF	0603HP-24N	
C_LPF1,C_LPF7	2	7.5 pF C0G 50 V		0402	TX LPF		
C_LPF3,C_LPF5	2	13 pF C0G 50 V		0402	TX LPF		
L_BPF1,L_BPF2	2	12 nH air-core 4T 1 mm wire on 3 mm (or dielectric 2-pole 439 MHz filter FL1)			BPF resonators 439 MHz / 50 MHz wide	commercial 430-450 MHz dielectric filter	handles 1.6 A circulating current

ref	qty	part	value_or_type	package	function	alternative	notes
C_BPF_E1,C_BPF_E2	2	2.7 pF C0G		0402	BPF end coupling		
C_BPF_C	1	1.0 pF C0G (0.5 + 0.5 pads)		0402	BPF coupling		
C_BPF_R1,C_BPF_R2	2	6.8 pF C0G + 1-3 pF trimmer pads		0402	BPF resonator		
L_M1,C_M1,C_M2	3	pi-match: shunt L / series C / shunt C (values from VNA)		0402/0603	antenna match		start 15 nH / 10 pF / NP
ANT1	1	normal-mode helix 60 mm 6 mm 16T on nylon former in TPU boot tuned 439 MHz			integral antenna 430-446 MHz (FM + LoRa 433)	PCB meander IFA (option B)	brass base insert + M3 boss
FL_CM1	1	90 ohm @100 MHz common-mode choke USB		0805	USB D+/D-		
F1	1	2S chemical fuse (Dexerials SFJ) 6 A			secondary protection		
Q1	1	dual N-FET 30 V 10 A low Rds		PowerPAK / TSSOP-8	battery protection switch		
BT1	2	Li-ion pouch 3.7 V 4000 mAh 606090-class 0.5C/2C		70x60x6 mm	battery 2S1P	2S2P 18650 (thicker variant)	UN 38.3 + IEC 62133-2 certified cells
NTC1-3	3	10 k NTC 1 %		0402 / leaded	cell / PA / charger temperature		
MAG1	1	Qi2 MPP / MagSafe ring magnet array N52 with orientation magnet and 0.3 mm steel shunt		OD ~56 mm	magnetic attachment	adhesive ring accessory for phones without one	centre 80 mm from top edge
SW1	1	tactile switch 12x6 mm side-actuated			PTT button		under TPU
SW2	1	tactile switch			power / mode		
LED1-2	2	bicolour LED		0603	battery / TX-RX		
CAB1	1	USB-C to USB-C 100 mm USB 2.0 3 A			phone cable		no e-marker needed
SHLD1-2	2	tin shield cans			RF area / power area		
Y1	1	8 MHz crystal (or 25 MHz for F407 variants)		3225	MCU clock; PLL to 168 MHz		check USB 48 MHz derivation

Appendix B. Schematic connection tables

Source: `hardware/schematics/`.

Sheet 1: USB-C, PD controller, charger, battery

J1 USB-C receptacle (16-pin)

Pin	Net	Notes
A4/B4, A9/B9 VBUS	VBUS	4 pins tied; 10 uF + 1 uH + 10 uF pi, SMBJ24A TVS
A5 CC1, B5 CC2	CC1, CC2	to TPS25751 CC1/CC2
A6/B6 D+, A7/B7 D-	USB_DP, USB_DM	both orientations tied; CM choke FL_CM1; TVS D1; 22 ohm to MCU
A8/B8 SBU	NC	
shell	GND via 1 nF	

U2 TPS25751D

Pin group	Net	Notes
CC1, CC2	CC1, CC2	
PP_HV / PP_EXT gate drive	VBUS source path FET Q2 (back-to-back N-FETs)	source path from BQ25798 OTG output `VBUS_OTG` to `VBUS`
Sink path	internal, VBUS to `VIN_CHG`	to BQ25798 VBUS pin
I2C_SDA/SCL, IRQ	I2C1_SDA, I2C1_SCL, PD_IRQ	address 0x21 (strap ADCIN)
GPIO0	CHG_OTG_EN	optionally directly enables BQ25798 OTG
LDO_3V3, VIN_3V3	+3V3D	
VBUS sense	VBUS	
Configuration	ADCIN1/ADCIN2 straps + I2C from MCU	DRP, Try.SRC, source PDOs 5V3A 9V2A (12V1.5A optional), sink PDOs 5V3A 9V3A 15V2A

U3 BQ25798

Pin	Net	Notes
VBUS	VIN_CHG	input from PD sink path
VAC1/VAC2	VIN_CHG	ACFET not used (TPS25751 handles)
SW1/SW2, BTST1/2	L1 2.2 uH 8 A	
SYS	VSYS	4 x 22 uF 25 V
BAT	VBAT+	through protector and 5 mOhm sense R_SNS
SDA/SCL/INT	I2C1, CHG_IRQ	address 0x6B
CE	CHG_CE (MCU)	pull-down enables charging
ILIM_HIZ	resistor to GND 1.5 A default; MCU overrides by I2C	
TS	NTC1 (cells) divider	0-45 C charge window
STAT	LED through MCU or direct	
OTG / QON	MCU GPIO	OTG request; QON for ship mode
REGN, PMID	per datasheet caps	PMID 2 x 22 uF
D+/D-	NC	BC1.2 detection unused (PD controller instead)

OTG output: `VBUS_OTG` = BQ25798 VBUS pin in OTG mode, routed through Q2 source-path FETs controlled by TPS25751.

Battery

Item	Connection
BT1 cells 2S	VBAT+ , CELL_MID, BAT-
U5 S-8252	VDD = VBAT+, VC = CELL_MID, VSS = BAT-, DO/CO drive Q1 dual N-FET in the BAT- return to GND; VM through 2 k
U6 BQ7718	senses each cell; OUT drives F1 chemical fuse heater between VBAT+ and the pack
U4 BQ34Z100-G1	BAT sense VBAT+ via divider, SRP/SRN across R_SNS 5 mOhm in the GND return, TS = NTC1, I2C address 0x55, ALERT to MCU
R_SNS	5 mOhm 2 W, in BAT- return before Q1

Balancing: BQ25798 has none; add a BQ29209 2S balancer (or a gauge with balancing) across CELL_MID. Marked as design option; cells matched at assembly.

Sheet 2: rails and MCU

U7 TPS62130 (+4V0, 3 A)

Pin	Net
VIN, PVIN	VSYS (10 uF + 100 nF)
EN	RADIO_RAIL_EN (MCU, pulled low at reset)
FSW	+3V3D (2.5 MHz) ; PG unused ; DEF = +3V3D forced PWM (check datasheet pin function)
SW	L2 1.0 uH 3 A
VOS/FB	divider for 4.00 V: R1 = 365 k, R2 = 100 k with 0.8 V reference -> V = 0.8 x (1 + 365/100) = 3.72... use R1 = 400 k, R2 = 100 k -> 4.00 V
OUT	+4V0_RAW -> L3 1 uH + 22 uF -> FB 600R -> +4V0 (100 uF + 100 nF + 1 nF at the module)

LDOs

Part	In	Out	Load
U8 TPS7A2033	+4V0	+3V3D	MCU, PD controller, gauge, LEDs, module UART level
U9 LP5907-3.3	+4V0	+3V3A	op-amps, mic bias, VREF+ of MCU

+3V3D is always on (needed to configure the PD controller); the MCU runs from it and enables the radio rail with `RADIO_RAIL_EN`.

U1 STM32F405RGT6 pin plan

Function	Pins	Net
USB FS	PA11 DM, PA12 DP	USB_DM, USB_DP (22 ohm series) ; PA9 VBUS sense through divider (20 V tolerant divider 10:1)
USART1 to module	PA9?? no: use USART2 PA2 TX, PA3 RX	RADIO_TX, RADIO_RX, 1 k + 100 pF RC at module
Module control	PB0 PTT (open-drain, 10 k pull-up to +3V3D), PB1 PD, PB2 H/L, PB10 SQ (EXTI, pull-up)	
ADC1_IN0 PA0	AUDIO_RX (from AA filter)	TIM6 trigger 16 kHz, DMA2
ADC2_IN1 PA1	MIC_LOCAL	same trigger
ADC1_IN4 PA4? reserved for DAC: DAC1_OUT1 = PA4	AUDIO_TX	DMA1, TIM6 trigger
ADC3_IN10 PC0, IN11 PC1	NTC_PA, NTC_BOARD	
I2C1 PB6 SCL, PB7 SDA	I2C1	2.2 k pull-ups to +3V3D

Function	Pins	Net
IRQs PB8 PD_IRQ, PB9 CHG_IRQ, PB12 GAUGE_ALERT	EXTI	
GPIO PC2 RADIO_RAIL_EN, PC3 SPK_EN, PC4 MIC_EN, PC5 CHG_CE, PC6 CHG_OTG, PC7 TEST_PORT_SENSE		TEST_PORT_SENSE: MM8130 has no switch sense; use a pogo/ground contact under the cover instead
Buttons PC8 PTT_BTN, PC9 PWR_BTN (wake-capable: use PA0 WKUP for PWR_BTN and move AUDIO_RX to PA5/ADC1_IN5)		
LEDs PC10, PC11		
SWD PA13/PA14, NRST, BOOT0	debug header	
HSE 8 MHz Y1	PLL -> 168 MHz core, 48 MHz USB	168 MHz x n never lands in 446.0-446.2 MHz (2.655x); 48 MHz x 9 = 432 MHz: keep clock traces short
VREF+	+3V3A via ferrite	
VDDA	+3V3A	

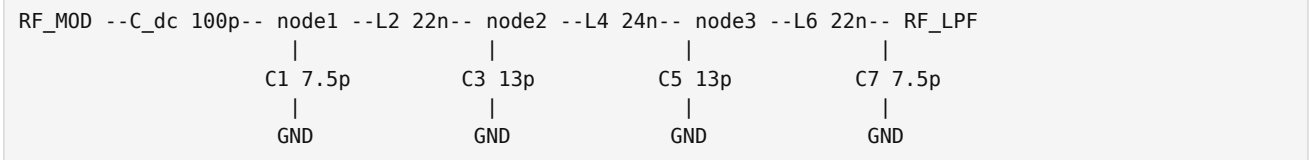
Decoupling 100 nF per VDD pin + 4.7 uF, VCAP 2 x 2.2 uF.

Sheet 3: radio module and RF chain

U10 SA818U-V4 (16-pin castellated; names per NiceRF datasheet, VERIFY)

Pin	Name	Connection
1	ANT	RF_MOD -> C_dc 100 pF -> LPF C1
2,3	GND	ground, many vias
4	PTT	PB0 through 1 k, 100 pF to GND; 10 k pull-up to +4V0 (inactive high)
5	PD	PB1 through 1 k (low = power down)
6	H/L	PB2 through 1 k (high = 1 W, low = 0.5 W; **VERIFY** polarity)
7	VCC	+4V0, 100 uF + 100 nF + 1 nF at pin
8	GND	
9	MIC_IN	from sheet 4 TX attenuator; 100 ohm + 1 nF at pin
10	GND	
11	AF_OUT	to sheet 4 RX buffer; 1 nF at pin
12	SQ	PB10 through 1 k; 100 pF; pull-up 10 k
13	NC	
14	RXD	RADIO_TX from MCU through 1 k
15	TXD	RADIO_RX to MCU through 1 k
16	NC / POW (SA868)	

LPF (values from hardware/rf/filter_design.py)



Each shunt capacitor has two ground vias; the inductors are placed in a line with 1 mm clearance, alternating orientation to reduce mutual coupling.

Optional BPF (option A footprint, bypass with 0 ohm 0402 links when unused)

```
RF_LPF --Ce 2.2p-- R1 --Cc 0.7p-- R2 --Ce 2.2p-- RF_BPF
                   |              |
             L 12n || C 6.8p + trimmer pads
                   |              |
                 GND            GND
```

Option B (RX-only SAW): two PE4259 SPDT footprints, control from PTT buffered by a 74LVC1G04; SAW footprint 3.0 x 3.0 mm; TX branch is a 0 ohm link.

Test switch, match, ESD, antenna

```
RF_BPF -> J2 MM8130-2600 (common in, antenna out, probe port) -> RF_TEST
RF_TEST --L_M1 shunt to GND (DC path / ESD) -- C_M1 series -- RF_ANT
                                                    |
                                               C_M2 shunt (NP by default)
RF_ANT -> D2 PESD0402-140 to GND -> antenna base pin ANT1
```

Guard ring of ground vias around the whole RF area; shield can SHLD1 over the module, LPF and BPF; the match and the test switch sit just outside the can so they can be probed.

Sheet 4: audio

RX: AF_OUT -> ADC and speaker

```
AF_OUT --1u (AC couple)--+-- U11A buffer (gain 2, biased 1.65 V from +3V3A divider)
                         |        |
                         |     Sallen-Key LPF 5 kHz: R 10k/10k, C 2.2n (to bias) / 4.7n (feedback)
                         |        |
                         |      AUDIO_RX -> PA0 ADC1_IN0 (1 k + 1 nF)
                         +-- U12 TPA6211A1 IN- (via 10 k, gain set 2), SHDN = SPK_EN  -> LS1 8 ohm
```

TX: DAC -> MIC_IN

```
DAC1_OUT1 (PA4) --> U13A Sallen-Key LPF 4 kHz (R 10k/10k, C 2.7n/5.6n)
                --> 1st-order HPF 300 Hz (C 100 nF, R 5.1 k to bias)
                --> divider 30:1 (R 15 k series, 510 ohm to GND) --> 100 ohm --> MIC_IN (1 nF at
module)
```

Divider ratio is a placeholder until the module's microphone sensitivity is measured (docs/07, T-R5c).

Local microphone

```
MK1 SPU0410LR5H: VDD = +3V3A via 100 ohm + 10 uF, OUT --1u--> U13B non-inverting gain 30 dB
  (R 1 k / 33 k, bias 1.65 V) --> 1st-order HPF 300 Hz --> MIC_LOCAL -> PA1 ADC2_IN1
MIC_EN gates the mic supply through a P-FET.
```

Grounding: AGND island under U11/U13/MK1 joined to GND at the MCU VSSA pin. Speaker amp returns to GND (not AGND).

Sheet 5: LoRa modules and antenna switch

U14 E22-400M22S (433 MHz) and U15 E22-900M22S (868 MHz)

Both modules have the same pinout family (**VERIFY** against the Ebyte datasheet revision): GND, VCC (3.3 V), NSS, SCK, MOSI, MISO, NRST, BUSY, DIO1, DIO2, RXEN, TXEN, ANT.

Module pin	U14 (433) net	U15 (868) net	Notes
VCC	+3V3R	+3V3R	10 uF + 100 nF at each module; TX peak 120 mA each
NSS	LORA433_NSS (PB12)	LORA868_NSS (PA15)	separate chip selects
SCK / MOSI / MISO	SPI2 PB13 / PB15 / PB14	same bus	8 MHz, 33 ohm series on SCK/MOSI
NRST	LORA433_RST (PC12)	LORA868_RST (PD2)	
BUSY	LORA433_BUSY (PB4)	LORA868_BUSY (PB5)	input
DIO1	LORA433_IRQ (PB3, EXTI)	LORA868_IRQ (PC13, EXTI)	TxDone / RxDone / Timeout
TXEN / RXEN	LORA433_TXEN / RXEN (PC14 / PC15)	LORA868_TXEN / RXEN (PA8 / PA10)	driven by the driver's rf_switch callback
DIO2	NC	NC	DIO2-as-switch not used
ANT	RF_L433 (to U16 SPDT RF2)	RF_L868 (to ANT2 match)	50 ohm CPW

U16 PE4259 SPDT (helix sharing)

Pin	Net
RFC	RF_SW_C -> BPF input (sheet 3)
RF1	RF_LPF (from the FM low-pass, sheet 3)
RF2	RF_L433 (from U14 ANT), with D4 BAP64-02 shunt PIN limiter footprint to GND through 100 pF
CTRL	ANT_SEL (PC11): low = FM, high = LoRa 433 (`hal_rf_antenna_to_lora433`)
VDD	+3V3R, 100 nF

DC blocks 100 pF on RF1/RF2/RFC. Layout: the switch sits between the FM LPF output and the BPF input inside the RF shield; keep RF1-RF2 isolation by grounding the gap (the 25 dB figure degrades quickly with coupling).

ANT2: 868 MHz printed inverted-F

```
RF_L868 --C 10 pF (series, tuning pad)--+-- IFA feed
                                        |
                                     L pad (shunt, NP default)
IFA short pin 4 mm from the feed to the ground plane edge
```

Keep-out 40 x 8 mm at the top-right PCB corner, no copper on any layer,

> = 30 mm from the helix base, >= 20 mm from the magnet ring. Ground plane edge straight along the element.

+3V3R

U17 TPS7A2033 from +4V0: 10 uF in/out, feeds U14, U15, U16 only.

Appendix C. Generated RF design tables

Output of `hardware/rf/filter_design.py` and `antenna_calc.py`.

Filter summary

Regenerate with `python3 hardware/rf/filter_design.py --out hardware/rf/results`.

```
TX harmonic LPF: 7th-order Chebyshev, 0.1 dB ripple, f_c = 500 MHz, Z0 = 50 ohm
g-values: 1.1812, 1.4228, 2.0967, 1.5734, 2.0967, 1.4228, 1.1812 (g_load = 1.0000)
element        ideal         E24
C1           7.52 pF     7.50 pF
L2          22.64 nH    22.00 nH
C3          13.35 pF    13.00 pF
L4          25.04 nH    24.00 nH
C5          13.35 pF    13.00 pF
L6          22.64 nH    22.00 nH
C7           7.52 pF     7.50 pF

LPF (ideal values, lossy parts)
-------------------------------
frequency                     S21 [dB]  S11 [dB]
PMR446 ch1 (446.006)              -0.8     -19.2
PMR446 ch16 (446.194)             -0.8     -19.2
TETRA uplink (380)                -0.5     -32.2
LPD433 / ISM (433.92)             -0.8     -18.6
LTE B28 UL (703)                 -36.7      -0.1
LTE B20 UL (832)                 -51.7      -0.1
2nd harmonic (892.2)             -57.8      -0.1
GSM900 UL (900)                  -58.5      -0.1
3rd harmonic (1338)              < -70      -0.0
4th harmonic (1784)              < -70      -0.0
5th harmonic (2230)              < -70      -0.0
Wi-Fi 2.4 GHz (2450)             < -70      -0.0
6th harmonic (2677)              < -70      -0.0

LPF (E24 values, lossy parts)
-----------------------------
frequency                     S21 [dB]  S11 [dB]
PMR446 ch1 (446.006)              -0.8     -16.4
PMR446 ch16 (446.194)             -0.8     -16.3
TETRA uplink (380)                -0.5     -26.6
LPD433 / ISM (433.92)             -0.7     -17.5
LTE B28 UL (703)                 -34.5      -0.2
LTE B20 UL (832)                 -49.8      -0.1
2nd harmonic (892.2)             -55.9      -0.1
GSM900 UL (900)                  -56.7      -0.1
3rd harmonic (1338)              < -70      -0.0
4th harmonic (1784)              < -70      -0.0
5th harmonic (2230)              < -70      -0.0
Wi-Fi 2.4 GHz (2450)             < -70      -0.0
6th harmonic (2677)              < -70      -0.0

RX/TX band-pass (optional): 2-pole Butterworth, f0 = 439.0 MHz, BW3dB = 50 MHz, L = 12.0 nH
  node C (bare resonator)        : 10.95 pF
  Qe in/out                      : 12.4 / 12.4
  end coupling caps Ce           : 2.70 / 2.70 pF
  coupling coefficient(s) k      : 0.0805
  coupling cap(s) Cc             : 0.96 pF
  resonator shunt caps (trimmed) : 7.62 pF, 7.62 pF
  est. insertion loss @ Qu=  45  : 2.40 dB
  est. insertion loss @ Qu= 120  : 0.90 dB
  est. insertion loss @ Qu= 300  : 0.36 dB

BPF alone (Qu = 120)
--------------------
frequency                     S21 [dB]  S11 [dB]
PMR446 ch1 (446.006)              -1.4     -17.6
PMR446 ch16 (446.194)             -1.4     -17.2
TETRA uplink (380)               -17.7      -0.3
LPD433 / ISM (433.92)             -1.2     -22.0
LTE B28 UL (703)                 -26.1      -0.1
LTE B20 UL (832)                 -28.3      -0.0
2nd harmonic (892.2)             -29.0      -0.0
GSM900 UL (900)                  -29.1      -0.0
```

```
3rd harmonic (1338)              -32.4      -0.0
4th harmonic (1784)              -36.7      -0.0
5th harmonic (2230)              -44.2      -0.0
Wi-Fi 2.4 GHz (2450)             -50.6      -0.0
6th harmonic (2677)              -62.7      -0.0

LPF(E24) + BPF cascade
----------------------
frequency                    S21 [dB]  S11 [dB]
PMR446 ch1 (446.006)             -2.0     -23.2
PMR446 ch16 (446.194)            -2.0     -23.4
TETRA uplink (380)               -18.5      -1.5
LPD433 / ISM (433.92)             -1.9     -18.0
LTE B28 UL (703)                 -66.0      -0.2
LTE B20 UL (832)                 < -70      -0.1
2nd harmonic (892.2)             < -70      -0.1
GSM900 UL (900)                  < -70      -0.1
3rd harmonic (1338)              < -70      -0.0
4th harmonic (1784)              < -70      -0.0
5th harmonic (2230)              < -70      -0.0
Wi-Fi 2.4 GHz (2450)             < -70      -0.0
6th harmonic (2677)              < -70      -0.0

Stop-band values below -70 dB are shown as a floor: on a real PCB, coupling around the filter (ground
return, pad-to-pad, shield gaps) limits isolation to roughly that level.

TX current: 141 mA RMS in the 50-ohm line at 1 W; resonator circulating current ~ Q_L x that = 1242 mA
RMS (Q_L = 8.8). Use air-core or coaxial ceramic resonators, not 0603 chips.
```

Antenna report

```
PMR446 centre 446.1 MHz: lambda = 672 mm, lambda/4 = 168 mm, lambda/2 = 336 mm

1. Fixed stubby helical (baseline antenna, integral, in a TPU boot)
   D =   6 mm, axial =  60 mm -> wire ~302 mm, pitch 3.82 mm, 15.7 turns
   D =   8 mm, axial =  50 mm -> wire ~302 mm, pitch 4.21 mm, 11.9 turns
   D =   5 mm, axial =  80 mm -> wire ~302 mm, pitch 4.31 mm, 18.6 turns
   Expected gain on the pack: -3 to -6 dBd; feed impedance 15-30 ohm, needs L-match.

2. Internal meandered IFA along the top PCB edge (option, phase 2)
   keep-out 60 x 10 mm -> conductor ~151 mm, 10 vertical runs, pitch 6.0 mm (ok)
   keep-out 70 x 12 mm -> conductor ~151 mm, 7 vertical runs, pitch 10.0 mm (ok)
   Expected gain stacked on a phone: -4 to -8 dBd; strongly detuned by the phone body.

3. ERP budget (limit 0.5 W ERP = 27.0 dBm ERP)
     P_cond   loss  G [dBd]  ERP [dBm]  ERP [W]  verdict
      1.0 W   1.1 dB     -2.0       26.9     0.49  OK
      1.0 W   1.1 dB     -4.0       24.9     0.31  OK
      1.0 W   1.1 dB     -6.0       22.9     0.19  OK
      0.5 W   1.1 dB     -2.0       23.9     0.24  OK
      0.5 W   1.1 dB     -4.0       21.9     0.15  OK
      0.5 W   1.1 dB     -6.0       19.9     0.10  OK
   loss = 0.8 dB LPF + 0.3 dB traces/ESD; antenna gain must be measured, firmware then
   fixes the module power level so that ERP <= 0.5 W with margin.

4. Range estimate (ERP 0.3 W, RX gain -4 dBi, sensitivity -118 dBm, 10 dB fade, 3 dB body loss/side)
   allowed path loss = 125 dB
   free space / line of sight       n = 2.0  ->  94.2 km
   open rural, ground reflections   n = 2.7  ->  4.8 km
   suburban                         n = 3.2  ->  1.3 km
   dense urban                      n = 3.8  ->  415 m
   indoor, through floors           n = 4.5  ->  162 m
   plane-earth two-ray, h = 1.5 m            ->  2.0 km

5. Example match: helix measured as 22 - j15 ohm at 446 MHz
   series(load)-shunt(source)   series  14.21 nH, shunt   8.05 pF, Q = 1.13
   series(load)-shunt(source)   series  36.33 pF, shunt  15.81 nH, Q = 1.13
```